# Self-Supervised Graph Masked Autoencoders for Hyperspectral Image Classification

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***Abstract*— Traditional supervised deep learning (DL) methods for hyperspectral image (HSI) classification are severely limited by the quality and quantity of labels. Furthermore, existing feature extraction methods generally lack the fusion of multiscale feature information, struggling to handle complex scenarios. To counter these problems, this work investigates a feature extraction module based on self-supervised graph masked autoencoders (SGMAEs). It innovatively employs graph masked autoencoders to achieve self-supervised label-free feature extraction for the complete set of samples, utilizing a multiscale graph convolutional network encoder (MGCNE) and cross correlation decoder (CCD) to extract and fuse multiscale spatial–spectral features of HSI data, respectively. Specifically, the HSI data is first converted into an edge-masked perturbed graph to label-freely extract multiscale feature representations of all pixel samples, and then fed into the MGCNE to obtain multilayer feature vectors for the pixel nodes. To reconstruct the masked edges for the fusion of multiscale features, the CCD applies cross correlation calculations to the nodes of the true edges at the masked positions and the fake edges at the random positions. The contrastive learning loss function is proposed for training of the autoencoder, which calculates the loss for the existence estimates of edges generated by cross correlation calculations. The pretrained MGCNE possesses an efficient self-supervised multiscale spatial–spectral feature extraction capability, along with strong generalizability, which significantly improves the accuracy of various mainstream models in downstream classification tasks. Extensive experiments and analyses on multiple HSI datasets demonstrate that our proposed SGMAE significantly enhances the model performance of various supervised classifiers and achieves superior performance in comparison to mainstream models.**

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## I. Introduction

HYPERSPECTRAL image (HSI) classification, as an important task in the field of remote sensing, has garnered significant attention in recent years. Hyperspectral imaging technology offers rich spectral information, enabling us to capture the spectral characteristics of surface objects through detailed analysis of electromagnetic waves at different wavelengths [1], [2]. These characteristics are not only instrumental in identifying and classifying complex land cover types but also play a vital role in various application scenarios, including environmental monitoring, agricultural management, and urban planning [3], [4]. With the continuous advancement of HSI data acquisition technologies and the development of emerging methods such as machine learning (ML) and deep learning (DL), improving the accuracy and efficiency of HSI classification has become a focal point and challenge in current research [5], [6], [7], [8], [9].

In traditional ML, the support vector machine (SVM) plays a milestone role in HSI classification [10], [11], [12], [13], demonstrating exceptional performance while effectively addressing the challenges posed by high-dimensional data. By constructing a maximum margin hyperplane, SVM accurately separates different classes, enhancing classification accuracy and robustness. Significant breakthroughs have been achieved in deep neural networks (DNNs) in modeling the rich structures of large datasets. In recent years, many supervised learning-based DL methods have been explored and widely implemented in HSI classification tasks. With the development of methods like convolutional neural networks (CNNs) [14], [15], [16], residual networks (ResNets) [17], and Transformers [18], researchers have made significant progress in enhancing classification accuracy and processing capability. Chen et al. [19] employed a regularized deep feature extraction method to improve the classification performance of CNNs on HSI. Zhong et al. [20] designed an end-to-end spatial–spectral ResNet (SSRN) that continuously studies spectral features and discriminative features in spatial context through spectral and spatial residual blocks, thereby improving model performance and alleviating the accuracy drop common in DNNs. Roy et al. [21] proposed an attention mechanism-based adaptive spectral-spectral kernel improved ResNet (A2S2K-ResNet) to effectively extract spectral–spatial features for HSI classification.

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Sun et al. [22] introduced a spectral-space feature token transformer (SSFTT) that combines 3-D and 2-D convolutional layers to extract low-level features, utilizing the Transformer framework to capture spectral–spatial features and high-level semantic features. Tu et al. [23] presented a local semantic feature aggregation transformer (LSFAT) architecture that effectively enhances the representation capability and computational efficiency of multiscale features in HSI classification through neighborhood aggregation modules and reusable classification tokens. However, the performance of DNNs based on supervised learning is sensitive to the quantity and quality of labeled data. Labeling large-scale datasets in the field of remote sensing is labor-intensive and costly. Meanwhile, the training samples for feature extraction in traditional supervised learning models are limited to a small portion of the labeled regions, resulting in an inability to mine information from the entire scene. These limitations of traditional supervised learning paradigms significantly hinder the effectiveness of DL in practical applications.

Self-supervised learning (SSL) effectively addresses the issues of data labeling and human supervision, achieving feature extraction from unlabelled data [24], [25]. SSL methods outperform traditional supervised methods in many tasks by relying on objectives generated from the data itself during training [26]. By designing self-supervised tasks (such as reconstructing input or generating targets), models can obtain high-level data representations that can be transferred to actual supervised downstream tasks. Generative self-supervised methods learn data representations by reconstructing or generating input data, with autoencoders being the most typical example [27], [28]. An autoencoder consists of an encoder and a decoder, with the encoder transforming input data into latent vectors and the decoder reconstructing the input data from these latent vectors. The main task of the encoder is feature extraction, while the decoder computes self-supervised loss to train the autoencoder [29], [30]. Ma et al. [31] extracted the spectral–spatial information of HSI through an improved spatial-denoising autoencoder (SDAE) and combined it with collaborative representation classification and graph-cutting spatial regularization techniques to enhance classification robustness in small training sets. Zhou et al. [32] designed a contrastive denoising stacked autoencoder (CDSAE) that optimizes each hidden layer of the stacked autoencoder by introducing local Fisher discriminative regularization, with the objective of learning low-dimensional feature mappings, minimizing scattering among similar samples, and maximizing separation across dissimilar samples while progressively training effective classifiers. Zhang et al. [33] designed a dual graph autoencoder (DGAE) that constructs superpixel similarity graphs and spectral band similarity graphs, employing a dual graph convolution structure to extract spatial information and spectral correlations from HSI for more discriminative feature representation. Cao et al. [34] enhanced classification performance by combining contrastive learning and mask autoencoder structures using SSL for feature extraction and classification on HSI data lacking accurate labels. However, previous autoencoder models have performed poorly in multiscale feature extraction and fusion of spatial–spectral information, which encounter challenges of coping with complex scenarios. Furthermore, their feature extraction capability for enhancing downstream classification task performance is limited.

A self-supervised graph masked autoencoders (SGMAEs) model for HSI classification is proposed to overcome these issues. As illustrated in Fig. 1, it consists of a multiscale GCN encoder (MGCNE) and a cross correlation decoder (CCD), which cascadingly form a parameter network of an autoencoder architecture, along with a contrastive learning loss function for SSL training of the autoencoder network. SGMAE can be used to enhance the model performance in downstream classification tasks. Specifically, the HSI data is converted into a perturbed graph by applying edge masking, and then fed into the MGCNE to extract multilayer feature vectors for the pixel nodes. The CCD applies cross correlation calculations to the nodes of the true edges at the masked positions and the fake edges at the random positions, generating existence estimates for both real edges and fake edges. These estimates are then used to calculate positive loss and negative loss, respectively, forming the contrastive learning loss function, which facilitates the training of the parameters of the encoder and decoder networks. Finally, the graph obtained from the HSI data is fed into the pretrained MGCNE to extract feature data for each sample in the entire HSI data, serving as input data for downstream classification tasks. SGMAE possesses an efficient multiscale spatial–spectral feature extraction capability along with strong generalizability, significantly improving the accuracy of various mainstream models in downstream classification tasks. The main contributions are summarized as follows.

1) An SSL-based SGMAE is proposed, which achieves label-free feature extraction from the complete set of HSI pixel samples, enhancing the performance of various mainstream models in downstream classification tasks.
2) An MGCNE is proposed, which efficiently achieves multiscale feature extraction for HSI data by constructing multilayer receptive fields of different scales.
3) A CCD is proposed, which innovatively utilizes cross correlation calculations on multilayer vectors of different receptive fields to reconstruct masked edges, thus efficiently achieving multiscale feature fusion and representation for HSI data.
4) The effectiveness of the proposed MGCNE and CCD is investigated and demonstrated. Extensive experiments show that our SGMAE outperforms several SOTA methods on four public HSI datasets.

The remainder of this article can be summarized as follows. Section II reviews previous related work. Section III introduces the proposed SGMAE. Section IV presents experiments and analyses. Section V summarizes the article and suggests future research directions.

## II. Related Work

### A. GNN for HSI Classification

In recent years, many graph neural networks (GNNs)-based methods [35] have been applied to HSI classification tasks, achieving positive results. Liu et al. [36] proposed a CNN-enhanced GCN (CEGCN) heterogeneous deep network that combines the characteristics of CNNs and GCNs for feature learning in smallscale regular areas and large-scale

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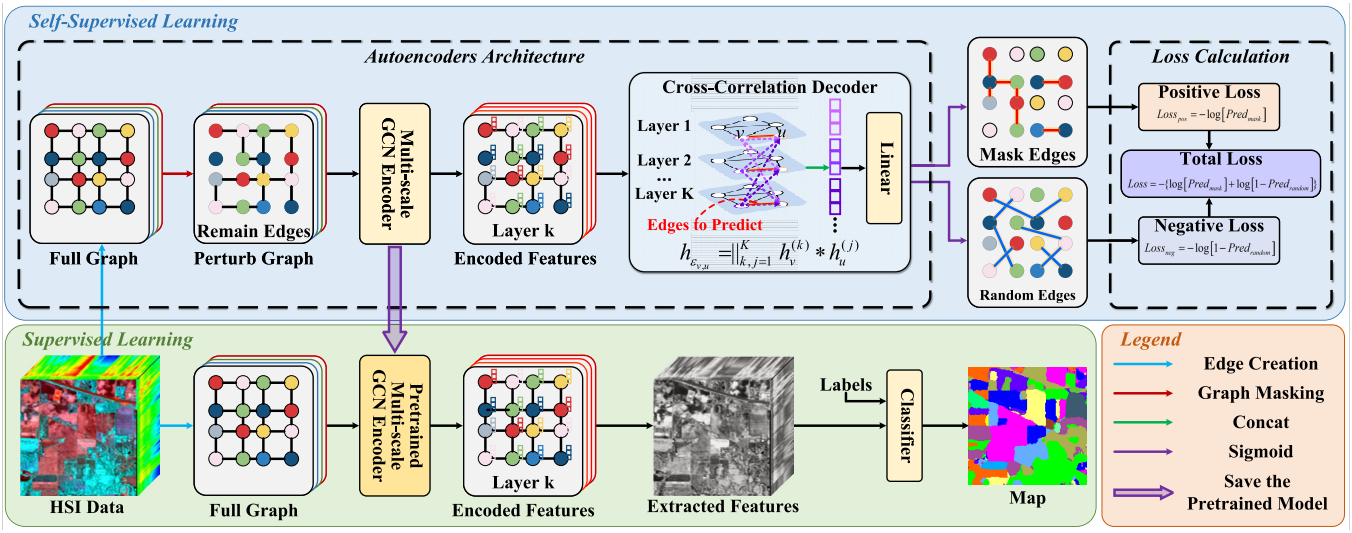



Fig. 1. Schematic view of the SGMAE architecture. It includes unsupervised autoencoder training and supervised learning for downstream classification tasks. In the unsupervised learning phase of the autoencoder architecture, HSI data are integrated into the autoencoder network parameter training in the form of a graph, undergoing masking, multiscale GCN encoding, and cross correlation decoding, reconstructing into masked edges and random edges. Subsequently, the network parameters of the autoencoder are trained using loss calculation based on contrastive learning. The pretrained encoder can then be used for feature extraction in downstream classification tasks, improving the performance of the classification model.

irregular regions. Yang et al. [37] introduced GraphMamba, an efficient HSI classification framework that integrates spatial–spectral feature preservation and computational efficiency through novel graph structure learning and adaptive spatial context awareness. Wang et al. [38] proposed an improved HGCN-MHF that enhances HSI classification under limited samples by integrating multihyperedge fusion and CNN feature extraction. Jamali et al. [39] proposed the development of an attention graph convolutional network (Attention-GCN) to address the shortcomings of CNNs in HSI classification, particularly the blurred classification boundary zone and the inability to capture global relationships. However, the training and feature extraction in these supervised learning methods are still limited to labeled data samples, and the performance of the model is seriously affected by the quality, quantity, and spatial distribution of the labeled data. It is difficult to remove the influence of labels and deeply mine features from the data itself.

### B. Graph-Based Semi/SSL for HSI Classification

The introduction of semi/SSL methods greatly leverages the advantages of feature extraction and information mining from unlabeled data. Xi et al. [40] utilized a cross-scale graph prototype network (X-GPN) framework to construct adjacency matrices for multiscale neighborhoods and integrated self-attention mechanisms to enhance the extraction of spectral–spatial features and classification accuracy, addressing issues of sample insufficiency and overfitting in HSI classification. Yang et al. [41] built a dynamic multiscale GCN (DMSGer), incorporating region-level superpixel segmentation and dynamic pixel feature updating strategies to capture complex relationships between pixels and regions in a semi-supervised manner. Jang et al. [42] proposed a new graph generative structure-aware Transformer (GraphGST) to address the position encoding issue in HSI classification by establishing a novel absolute position encoding (APE) integrated into the Transformer architecture, along with a generative mechanism leveraging SSL for cross-view contrastive learning. Chang et al. [43] proposed an unsupervised multiview graph contrastive (UMGC) feature learning algorithm for HSI classification, which explores deep semantic features without being constrained by samples by constructing multiview adjacency matrices, enhancing graph topology and attribute structures through adaptive data augmentation, and maximizing feature similarity between views using a contrastive loss. Compared to these semi-supervised/self-supervised feature extraction methods, our proposed approach enables efficient extraction and fusion of multiscale spatial–spectral features, and it exhibits strong generalization capability across various mainstream models in downstream classification tasks, significantly improving the performance of these classification models.

## III. Proposed Approach

The proposed SGMAE is outlined in Fig. 1. It first transforms the HSI data into graph data and applies edge masking to construct a perturbed graph. It also includes an MGCNE capable of label-free feature extraction for pixel-wise samples and a CCD that enables multiscale feature fusion. The reconstructed edges utilize a contrastive learning loss function for loss calculation, thus training the autoencoder (MGCNE and CCD). The pretrained SGMAE can enhance the performance of classifiers for downstream classification tasks.

### A. Adjacency Edge Creation to Construct Graph

HSI data is a 3-D data cube where each pixel is a high-dimensional spectral feature vector. The pixel points conform to the rules of Euclidean geometric arrangement, meaning they have relative reference anchor points, a sequence, and determined relationships in terms of Euclidean distance. Compared to the sample arrangement structure of a matrix, the graph structure offers higher spatial efficiency

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**Algorithm 1** Self-Supervised Feature Extraction in SGMAE

**Input:** $X : (H, W, C)$
**Output:** $F : (H, W, D)$

$X_{pca} : (H, W, C_{pca}) \leftarrow PCA(X)$
$E_{full} \leftarrow EdgeConstruction(X_{pca})$
$E_{train}, E_{val} \leftarrow EdgeSplit(E_{full})$
$E_{val_pos} \leftarrow E_{val}$
$E_{val_neg} \leftarrow NegativeSampling(E_{val})$
$Adj_{full} \leftarrow DoSparse(E_{full})$
$E_{train_remain}, E_{train_mask} \leftarrow Mask(E_{train})$
$Adj_{train_remain} \leftarrow DoSparse(E_{train_remain})$
**for** each $i \in epoch$ **do**
  $h : (H, W, D) \leftarrow Encoder(Adj_{train_remain}, X_{pca})$
  $E_{train_mask_pred} \leftarrow Decoder(E_{train_mask}, h)$
  $loss_{pos} \leftarrow -\log(E_{train_mask_pred})$
  $E_{train_random} \leftarrow Random(N_{nodes}, N_{edges})$
  $E_{train_random_pred} \leftarrow Decoder(E_{train_random}, h)$
  $loss_{neg} \leftarrow -\log(1 - E_{train_random_pred})$
  $loss \leftarrow loss_{pos} + loss_{neg}$
  $Encoder_{Trained}(), Decoder_{Trained}() \leftarrow backward$
  $h' : (H, W, D) \leftarrow Encoder_{Trained}(Adj_{full}, X_{pca})$
  $E_{val_pos_pred} \leftarrow Decoder_{Trained}(E_{val_pos}, h')$
  $E_{val_neg_pred} \leftarrow Decoder_{Trained}(E_{val_neg}, h')$
  $E_{val_pred} \leftarrow Concat(E_{val_pos_pred}, E_{val_neg_pred})$
  $E_{val_true} \leftarrow Concat([1, \ldots, 1], [0, \ldots, 0])$
  $auc \leftarrow RocAucScore(E_{val_pred}, E_{val_true})$
  **if** $auc ==$ max$(each\ auc)$ **then**
    $Encoder_{FinalTrained}() \leftarrow Save$
  **end if**
**end for**
$F : (H, W, D) \leftarrow Encoder_{FinalTrained}(Adj_{full}, X_{pca})$
**return** $F$

as well as the ability to express more complex and precise relationships between pixel nodes, which is particularly emphasized when using a random masking strategy. To facilitate subsequent graph masking, graph convolution, and graph generation of the HSI data, it is necessary to first convert the 3-D cube HSI data into graph data. Here, a pixel-wise sample adjacency edge construction method is adopted to build the edges in the graph data. Each pixel point is traversed, and association edges are constructed among pixels within a neighborhood centered around it, such as $3 \times 3$ or $5 \times 5$, using four-adjacency, eight-adjacency, or 24-adjacency. This approach allows for the spatial information interaction among neighboring pixels.

Then, use principal component analysis (PCA) [44] to reduce the dimensionality of HSI $\mathcal{X} \in \mathbb{R}^{H \times W \times C}$ with a height of $H$, a width of $W$, and $C$ channels to $P$, in order to eliminate information redundancy in the node spectral vectors, as shown in the formula

$$\mathcal{X}_{\text{pca}} = \text{PCA}(\mathcal{X}, P) \tag{1}$$

where PCA$(\cdot)$ is the principal component analysis (PCA) dimension reduction operation, $\mathcal{X}$ is the original HSI data, $P$ is the number of channels in the dimension-reduced image data, and the dimension-reduced image data is $\mathcal{X}_{\text{pca}} \in \mathbb{R}^{H \times W \times P}$.

Each pixel of the dimension-reduced image data $\mathcal{X}_{\text{pca}}$ is treated as a node $v_i$, for $i = 1, 2, \ldots, H \times W$ of the graph, and the complete set of nodes formed by all pixel points is represented as $\mathcal{V}$, as shown in the formula

$$\mathcal{V} = \left\{ v_i | v_i \in \mathbb{R}^P \right\}_{i=1}^{H \times W}. \tag{2}$$

Each node $v_i \in \mathbb{R}^P$ is a high-dimensional feature vector of dimension $P$, and then nodes are connected to each other if their Euclidean distance in the coordinates is not greater than $D$. The connected edges form the complete set of adjacency edges $\varepsilon$, and the constructed edge set is as shown in the formula

$$\varepsilon = \left\{ \left(v_i, v_j\right) | \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2} \leq D \right\}_{i=1, j=1}^{H \times W} \tag{3}$$

where $D$ represents the set Euclidean distance range, typically set to 1 or 2. $x_i$ and $y_i$ are the horizontal and vertical coordinates of the source node $v_i$, while $x_j$ and $y_j$ are the horizontal and vertical coordinates of the destination node $v_j$. The feature vectors of all nodes form the node feature matrix $F \in \mathbb{R}^{|\mathcal{V}| \times P}$, where $|\mathcal{V}| = H \times W$ and $|\mathcal{V}|$ represent the number of nodes. The final created graph data is $\mathcal{G} = (\mathcal{V}, \varepsilon)$, thus completing the conversion from HSI $\mathcal{X}$ to graph data $\mathcal{G}$.

### B. Multiscale GCN Encoder

After constructing the graph data derived from a 3-D data cube, the graph still exhibits the unique spatial rules and fixed relationships characteristic of a matrix, which may result in weaker generalization during model training. Therefore, specific data augmentation methods, namely, random edge masking, need to be adopted to create more complex and generalized graph structure information. This approach helps the encoder leverage GNNs to facilitate deep information interaction among node features, learn the high-level features of the data, enrich the model's representational capability, and mitigate the pattern rigidity caused by spatial positional relationships.

The steps are as follows to create the masked edge set $\varepsilon_{\text{mask}}$ and the remain edge set $\varepsilon_{\text{remain}}$ using random edge masking. First, set a certain masking ratio $m$, and then apply the random edge masking to the full edge set $\varepsilon$, as shown in the formula

$$\varepsilon_{\text{mask}} = \text{mask}(\varepsilon, m) \tag{4}$$

where $\varepsilon_{\text{mask}}$ represents the set of masked edges, and mask$(\cdot)$ refers to the random edge masking operation on the graph. Among them, $\varepsilon_{\text{mask}} \subset \varepsilon$ while $\varepsilon_{\text{remain}} = \varepsilon - \varepsilon_{\text{mask}}$, which are used to obtain the remain edge set $\varepsilon_{\text{remain}}$ that are not masked. The remain edge set $\varepsilon_{\text{remain}}$ combined with the full set of nodes $\mathcal{V}$ form the perturbed graph $\mathcal{G}_{\text{perb}} = (\mathcal{V}, \varepsilon_{\text{remain}})$, which is the graph after masking.

In the following, a loss function based on contrastive learning is designed, and a quantity of $\varepsilon_{\text{random}}$ equal to that of $\varepsilon_{\text{mask}}$ needs to be constructed, treating $\varepsilon_{\text{mask}}$ as the positive sample and $\varepsilon_{\text{random}}$ as the negative sample. The construction method of $\varepsilon_{\text{random}}$ is given by the following formula:

$$\begin{aligned} \varepsilon_{\text{random}} &= \left\{ (v_i, v_j) | i, j \sim \text{Uniform}(1, H \times W), i \neq j \right\} \\ \varepsilon_{\text{random}} &\cap \varepsilon_{\text{mask}} = \emptyset, \quad |\varepsilon_{\text{random}}| = |\varepsilon_{\text{mask}}| \end{aligned} \tag{5}$$

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where Uniform$(\cdot)$ indicates a function that generates random numbers uniformly distributed over a specified interval. The above expression indicates that the two sets $\varepsilon_{\text{mask}}$ and $\varepsilon_{\text{random}}$ do not have any identical edges, and the number of edges is the same.

After completing a series of operations for graph masking, the perturbed graph data $\mathcal{G}_{\text{perb}} = (\mathcal{V}, \varepsilon_{\text{remain}})$ is fed into a multilayer GCN encoder $GCN_k(\cdot)$, yielding a hidden vector representation $h^{(k)} \in \mathbb{R}^{|\mathcal{V}|\times d}$ of nodes at each GCN layer, obtaining a total of $K$ layers of node feature vector representations. Feed the perturbed graph, which uses the spectral feature vectors $F \in \mathbb{R}^{|\mathcal{V}|\times P}$ of the graph nodes and the remaining edge set $\varepsilon_{\text{remain}}$ into the GCN [45] encoder, as represented in the formula

$$\begin{aligned} h^{(1)} &= \text{GCN}_1(F, \varepsilon_{\text{remain}}) \\ h^{(k)} &= \text{GCN}_k\left(h^{(k-1)}, \varepsilon_{\text{remain}}\right), \quad \text{for } k = 2, 3, \ldots, K \end{aligned} \tag{6}$$

where $\text{GCN}_k(\cdot)$, for $k = 1, 2, \ldots, K$ denotes the GCN layer at the $k$th layer, $h^{(k)}$, for $k = 1, 2, \ldots, K$ represents the hidden vector of the nodes at the $k$th layer encoded by the $k$th layer of the GCN, and $K$ indicates the specified number of GCN layers. $h^{(k)} \in \mathbb{R}^{|\mathcal{V}|\times d}$, $|\mathcal{V}|$ are the number of nodes, while $d$ represents the dimensionality of the hidden vectors. $h$ refers to the encoded feature vector obtained after encoding the spectral feature vectors through MGCNE.

### C. Cross Correlation Decoder

After the encoding of the node feature vectors, they need to be decoded. Tan et al. [30] using the feature vectors of the nodes at both ends of the edge to be predicted, a feature vector representation for this edge is generated, and then the probability of its existence is output.

To generate the vector representation of the edge to be predicted $\varepsilon_{v,u}$, the source and destination nodes are denoted as $v$ and $u$, respectively. Their node encoding feature vectors at each layer are represented as $\{h_v^{(k)}, h_u^{(k)}\}_{k=1}^K$. These vectors are input into a CCD to obtain the vector representation $h_{\varepsilon_{v,u}}$ of the edge to be predicted $\varepsilon_{v,u}$ between the source and destination nodes, as shown in the formula

$$h_{\varepsilon_{v,u}} = \|_{k,j=1}^K h_v^{(k)} * h_u^{(j)} \tag{7}$$

where $*$ denotes the element-wise multiplication, and $\|$ denotes the vector concatenation. The cross correlation calculation involves performing element-wise multiplication between the hidden node vectors $h_v^{(k)}$, for $k = 1, 2, \ldots, K$ of each layer for the source node $v$ and the hidden node vectors $h_u^{(k)}$, for $k = 1, 2, \ldots, K$ of each layer for the destination node $u$. Then, the resulting $K^2$ vectors are concatenated along the spectral dimension to obtain the vector representation $h_{\varepsilon_{v,u}}$ of the edge to be predicted $\varepsilon_{v,u}$. The $\varepsilon_{\text{mask}}$ and $\varepsilon_{\text{random}}$ to be predicted are fed into the decoder and the vectors representation of these edges are obtained, as shown in formulae

$$h_{\varepsilon_{\text{mask}}} = \|_{k,j=1}^K h_v^{(k)} * h_u^{(j)}, \quad \varepsilon_{\text{mask}} = \{(v, u)\} \tag{8}$$

$$h_{\varepsilon_{\text{random}}} = \|_{k,j=1}^K h_v^{(k)} * h_u^{(j)}, \quad \varepsilon_{\text{random}} = \{(v, u)\} \tag{9}$$

where $\varepsilon_{\text{mask}} = \{(v, u)\}$ indicates that the two end nodes of any edge in the edge set are $v$ and $u$. $h_{\varepsilon_{\text{mask}}}$ and $h_{\varepsilon_{\text{random}}}$ are the vector representations of $\varepsilon_{\text{mask}}$ and $\varepsilon_{\text{random}}$, respectively.

Finally, the feature vectors $h_{\varepsilon_{\text{mask}}}$ and $h_{\varepsilon_{\text{random}}}$ of $\varepsilon_{\text{mask}}$ and $\varepsilon_{\text{random}}$ are put into the multilayer perceptron (MLP) layer and Sigmoid layer to obtain the final existence prediction probability, as shown in formulae

$$\begin{aligned} \text{Pred}_{\varepsilon_{\text{mask}}} &= \text{Sigmoid}[\text{MLP}(h_{\varepsilon_{\text{mask}}})] \\ \text{Pred}_{\varepsilon_{\text{mask}}} &= \{p_{\text{mask}} | 0 < p_{\text{mask}} < 1\} \end{aligned} \tag{10}$$

$$\begin{aligned} \text{Pred}_{\varepsilon_{\text{random}}} &= \text{Sigmoid}[\text{MLP}(h_{\varepsilon_{\text{random}}})] \\ \text{Pred}_{\varepsilon_{\text{random}}} &= \{p_{\text{random}} | 0 < p_{\text{random}} < 1\} \end{aligned} \tag{11}$$

where MLP$(\cdot)$ represents the MLP layer and Sigmoid$(\cdot)$ represents the Sigmoid layer. $\text{Pred}_{\varepsilon_{\text{mask}}}$ and $\text{Pred}_{\varepsilon_{\text{random}}}$ represent the prediction probabilities of all edges in the $\varepsilon_{\text{mask}}$ set and $\varepsilon_{\text{random}}$ set, respectively. $p_{\text{mask}}$ and $p_{\text{random}}$ represent the prediction probability of any edge in the $\varepsilon_{\text{mask}}$ set and $\varepsilon_{\text{random}}$ set, respectively, both of which are between 0 and 1.

### D. Loss Function Based on Contrastive Learning

A loss function based on contrastive learning has been designed, which allows the prediction result of $\varepsilon_{\text{mask}}$ be 1, indicating the existence, while the prediction result of $\varepsilon_{\text{random}}$ is to be 0, indicating nonexistence. This is obvious because $\varepsilon_{\text{mask}}$ is real when it is first constructed, and it can truly reflect the spatial position relationship between adjacent nodes. $\varepsilon_{\text{random}}$ is randomly selected, it is not a real edge, and it cannot reflect the spatial position relationship between adjacent nodes. The loss function is as shown in the formula

$$\text{Loss} = -[\log \text{Pred}_{\text{mask}} + \log(1 - \text{Pred}_{\text{random}})] \tag{12}$$

where $-\log \text{Pred}_{\text{mask}}$ is the positive loss, and $-\log(1 - \text{Pred}_{\text{random}})$ is the negative loss. They are two independent loss calculation processes, and therefore two independent cross-entropy loss functions applied to binary classification. The general binary classification cross-entropy loss function can be written as the formula

$$\text{Loss}_{\text{cross}} = -[Y \log P + (1 - Y) \log(1 - P)] \tag{13}$$

where $Y$ is the labels of samples, with the positive class being 1 and the negative class being 0, and $P$ is the probability that the model predicts the samples to be positive. For the positive loss, all labels of the samples are 1, and $\text{Pred}_{\text{mask}}$ is equal to $P$. The positive loss function in the form of a cross-entropy loss function can be written as the formula

$$\text{Loss}_{\text{cross-pos}} = -\log \text{Pred}_{\text{mask}}. \tag{14}$$

For the negative loss, all labels of the samples are 0, and $\text{Pred}_{\text{random}}$ is equal to $P$. The negative loss function in the form of a cross-entropy loss function can be written as the formula

$$\text{Loss}_{\text{cross-neg}} = -\log(1 - \text{Pred}_{\text{random}}). \tag{15}$$

The cross-entropy form of the total loss function, which combines the positive loss function and the negative loss function, can be written as shown in (12), as the formula

$$\begin{aligned} \text{Loss} &= \text{Loss}_{\text{cross-total}} \\ &= \text{Loss}_{\text{cross-pos}} + \text{Loss}_{\text{cross-neg}} \\ &= -[\log \text{Pred}_{\text{mask}} + \log(1 - \text{Pred}_{\text{random}})]. \end{aligned} \tag{16}$$

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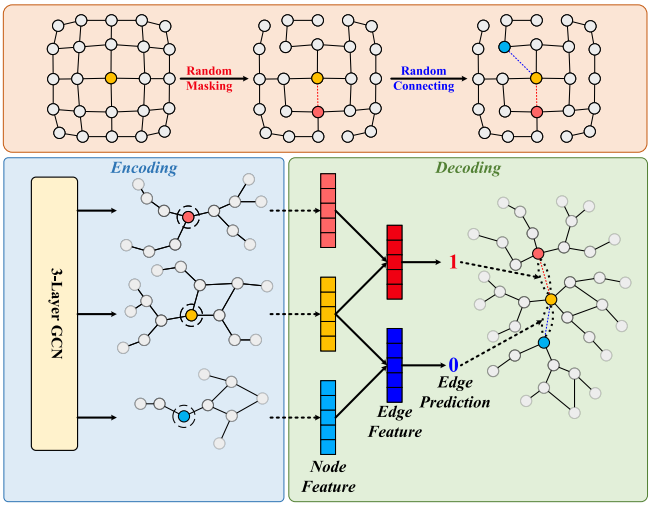



Fig. 2. Detailed mechanism diagram of the encoding and decoding process.

As the Loss decreases, $\text{Pred}_{\text{mask}}$ gradually increases and $\text{Pred}_{\text{random}}$ gradually decreases.

Repeat the training for multiple epochs to achieve the training of both MGCNE and CCD, and adjust the network parameters. When the loss is reduced to a sufficiently small value, the expected output $p_{\text{mask}}$ is 1 and $p_{\text{random}}$ is 0, as shown in the following formula:

$$\lim_{\text{Loss}\to 0} p_{\text{mask}} = 1$$
$$\lim_{\text{Loss}\to 0} p_{\text{random}} = 0. \tag{17}$$

### E. SGMAE for Downstream Classification Tasks

Input the original graph data $\mathcal{G} = (\mathcal{V}, \varepsilon)$ into the pretrained MGCNE $\text{GCN}_k(\cdot)$, for $k = 1, 2, \ldots, K$ for feature extraction of image pixel samples, resulting in the feature vector of the samples $f_i \in \mathbb{R}^d$, for $i = 1, 2, \ldots, |\mathcal{V}|$, which forms the image feature matrix $F \in \mathbb{R}^{W\times H\times d}$. Typically, take $f_i = h_i^{(k)}$ or $f_i = \|_k^K h_i^{(k)}$, where usually let $k = K$ or concatenate the encoded feature vectors from all layers. The encoded image feature matrix $F \in \mathbb{R}^{W\times H\times d}$ is then directly input into commonly used supervised learning classifiers (e.g., CNN) for training to obtain classification results, achieving improvements in classification performance.

### F. Algorithm Details

The specific code implementation details of the SGMAE feature extraction module proposed are shown in Algorithm 1. The overall framework is shown in Fig. 1 and its subfigures Figs. 2 and 3.

Fig. 2 illustrates some details about the encoding and decoding processes in the main framework diagram. In the light orange box, an HSI data is first constructed into a graph. In this graph, only the central node is taken as the study object, while irrelevant data are omitted. The central node is depicted in orange. Initially, the entire graph undergoes masking, which obscures some edges. However, the masked edges connected to the study node are highlighted as red dashed lines. Subsequently, some random edges are generated, as described in (5), and only the random edges connected to the study node are drawn as blue dashed lines. In the blue box, the study node and the two nodes connected to it via masked edges and random edges are shown. After being processed through multiple layers of GCN encoding, they generate feature nodes that aggregate information from multiorder neighboring nodes. These nodes point to the feature vectors of the three nodes represented in the green box. The feature vector of the study node is combined with the vectors of the other two nodes to compute a new vector, representing the edge feature vectors between the central node and these two nodes. Their output is a real number, serving as the predicted probability for the existence of the edge. After sufficient training, the model can predict the masked edges with an output of 1 and the random edges with an output of 0.

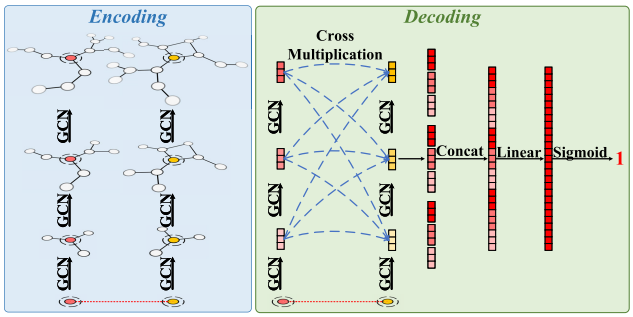



Fig. 3. Detailed mechanism diagram of the CCD.

Fig. 3 illustrates more details of the CCD. In the blue box of the left figure, the extension of high-order neighboring nodes during the three-layer GCN encoding process for each node is represented, which also implies that the receptive field of high-order features continuously expands as the number of convolutional layers increases. In the green box of the right figure, corresponding to the left figure, the node feature vectors output by these two nodes after three layers of GCN are displayed. These vectors are cross-correlated according to (7), generating a series of vectors that are then concatenated and output to predict whether the edge between these two nodes is a masked edge.

## IV. Experiments

### A. Datasets Description

The section will report on classification experiments conducted on four HSI datasets, including Indian Pines, Pavia University, Salinas, and Houston 2018. Relevant information about these datasets will be presented in detail next.

*1) Indian Pines Dataset:* The Indian Pines dataset was acquired in 1992 by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) in the Indian Pines region of Indiana, USA. The data size is 145 × 145 pixels, and a total of 10 249-pixel samples are labeled for training and testing purposes in HSI classification. The wavelength range of the AVIRIS spectrometer is from 0.4 to 2.5 m, capable of imaging the ground over 220 consecutive bands; 20 bands that could not be reflected by water were removed, leaving a total of 200 bands for analysis.

*2) Pavia University Dataset:* The Pavia University dataset was acquired in 2003 by the Reflective Optics Spectrographic Imaging System (ROSIS-03) in parts of the city of Pavia, Italy. The data size is 610 × 340 pixels, with 42 776-pixel samples

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TABLE I

NAME AND NUMBER OF SAMPLES OF EACH CLASS IN FOUR DATASETS. INDIAN PINES HAS 10% LABELED DATA. PAVIA UNIVERSITY HAS 1% LABELED DATA. SALINAS HAS 0.5% LABELED DATA. HOUSTON 2018 HAS 5% LABELED DATA

| | Indian Pines | | | Pavia University | | | Salinas | | | Houston 2018 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| No. | Class Name | Train | Test | Class Name | Train | Test | Class Name | Train | Test | Class Name | Train | Test |
| 1 | Alfalfa | 5 | 41 | Asphalt | 66 | 6565 | Brocoli_green_weeds_1 | 10 | 1999 | Healthy grass | 490 | 9309 |
| 2 | Corn-notill | 143 | 1285 | Meadows | 186 | 18463 | Brocoli_green_weeds_2 | 19 | 3707 | Stressed grass | 1625 | 30877 |
| 3 | Corn-mintill | 83 | 747 | Gravel | 21 | 2078 | Fallow | 10 | 1966 | Synthetic grass | 34 | 650 |
| 4 | Corn | 24 | 213 | Trees | 31 | 3033 | Fallow_rough_plow | 7 | 1387 | Evergreen trees | 680 | 12915 |
| 5 | Grass-pasture | 48 | 435 | Sheets | 13 | 1332 | Fallow_smooth | 13 | 2665 | Deciduous trees | 251 | 4770 |
| 6 | Grass-trees | 73 | 657 | Bare soil | 50 | 4979 | Stubble | 20 | 3939 | Soil | 226 | 4290 |
| 7 | pasture-mowed | 3 | 25 | Bitumen | 13 | 1317 | Celery | 18 | 3561 | Water | 13 | 253 |
| 8 | Hay-windrowed | 48 | 430 | Bricks | 37 | 3645 | Grapes_untrained | 56 | 11215 | Residential Bldg. | 1989 | 37783 |
| 9 | Oats | 2 | 18 | Shadows | 10 | 937 | Soil_vinyard_develop | 31 | 6172 | Non-residential Bldg. | 11187 | 212565 |
| 10 | Soybean-notill | 97 | 875 | | | | Corn_senesced_weeds | 16 | 3262 | Roads | 2293 | 43573 |
| 11 | Soybean-mintill | 245 | 2210 | | | | Lettuce_romaine_4wk | 5 | 1063 | Sidewalks | 1702 | 32327 |
| 12 | Soybean-clean | 59 | 534 | | | | Lettuce_romaine_5wk | 10 | 1917 | Crosswalks | 76 | 1442 |
| 13 | Wheat | 20 | 185 | | | | Lettuce_romaine_6wk | 5 | 911 | Major thoroughfares | 2317 | 44031 |
| 14 | Woods | 126 | 1139 | | | | Lettuce_romaine_7wk | 5 | 1065 | Highways | 493 | 9372 |
| 15 | Buildings etc. | 39 | 347 | | | | Vinyard_untrained | 36 | 7232 | Railways | 347 | 6590 |
| 16 | Stone etc. | 9 | 84 | | | | Vinyard_vertical_trellis | 9 | 1798 | Paved parking lots | 575 | 10925 |
| 17 | | | | | | | | | | Unpaved parking lots | 7 | 139 |
| 18 | | | | | | | | | | Cars | 327 | 6220 |
| 19 | | | | | | | | | | Trains | 269 | 5100 |
| 20 | | | | | | | | | | Stadium seats | 341 | 6483 |
| | Total | 1024 | 9225 | Total | 427 | 42349 | Total | 270 | 53859 | Total | 25242 | 479614 |

labeled for training and testing purposes in HSI classification. The wavelength range of the ROSIS-03 spectrometer is from 0.43 to 0.86 m, capable of imaging the ground over 115 consecutive bands. 12 bands severely affected by noise were removed, leaving a total of 103 bands for analysis.

*3) Salinas Dataset:* The Salinas dataset was acquired by the AVIRIS in the Salinas Valley, CA, USA. The data size is 512 × 217 pixels, with 54 129 pixel samples labeled for training and testing purposes in HSI classification. The image was continuously imaged over 224 bands, from which 20 bands that could not be reflected by water were removed, leaving a total of 204 bands for analysis.

*4) Houston 2018 Dataset:* The Houston 2018 dataset [46] was acquired by the HSI Analysis Group and the NSF-funded Airborne Laser Mapping Center (NCALM) at the University of Houston, covering the university campus and surrounding areas. The data size is 601 × 2384 pixels, with 504 856-pixel samples labeled for training and testing purposes in HSI classification. The wavelength range of the image data is from 0.38 to 1.05 m, with a total of 50 bands.

The partitioning of the training set and test set, along with the sample counts for each category, is shown in Table I.

### B. Experimental Setting

*1) Quality Indexes:* Unsupervised feature extractor for binary classification: during the unsupervised feature extraction training phase of SGMAE, model validation is set for each training epoch, and the AUC value for each epoch is calculated and plotted on the ROC curve.

Supervised classifier for multiclass classification: the average was calculated by conducting ten repeated experiments on randomly selected training set samples, resulting in three widely used quality metrics: overall accuracy (OA), average accuracy (AA), and Kappa coefficient ($\kappa$).

*2) Competitive Approaches:* Four representative classification methods were used for comparison.

*a) Conventional methods:* The classic SVM classifier was employed to process the data [10], [11], [12], [13]. First, feature scaling was applied to ensure standardized handling of the data. In terms of parameter settings, a grid search method was used to establish the parameter grid, operating with a radial basis function (RBF) kernel. The two hyperparameters and were optimized through fivefold cross-validation within the ranges $\sigma = [2^{-3}, 2^{-2}, \ldots, 2^4]$ and $\lambda = [2^{-2}, 2^{-1}, \ldots, 2^4]$. The CNN [14], [15], [47] consists of three convolutional layers and one linear layer, with each convolutional layer having a kernel size of 3 × 3 × 3. The number of kernels in the layers is 32, 64, and 128, respectively, and each layer uses the same size of padding, ReLU activation function, and 2 × 2 × 2 max pooling, with the final linear layer directly outputting classification results.

*b) Residual network:* The advanced methods based on ResNet, namely, SSRN [20] and attention-based adaptive spectral–spatial kernel ResNet (A2S2K) [21] are compared. Both of these models employ the concept of residual connections, which can prevent overfitting and maintain efficiency with deeper network parameters, achieving multiscale information fusion for HSI. They demonstrate high OA and computational time costs. SSRN and A2S2K are advanced models combining convolutional and attention mechanisms for feature extraction and classification. SSRN starts with Conv2d, LeakyReLU, and BatchNorm2d, followed by 3-D convolutional layers (SPC32) and SARes layers with spatial attention (SpatAttn) for feature recalibration. It uses Conv2d for query/key/value transformations, Softmax for attention weighting, and BatchNorm2d with ReLU for normalization, with additional Conv2d layers refining features. A2S2K begins with parallel Conv3d layers, batch normalization, ReLU, and AdaptiveAvgPool3d for dimension reduction. It includes a squeeze-and-excitation block and four residual blocks with Conv3d, batch normalization, ReLU, and ECA layers for feature refinement. Both models integrate convolutional and attention mechanisms for robust feature learning.

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*c) Transformer:* The advanced methods based on Transformer, namely, spectral–spatial transformer network (SSTN) [48], spectral–spatial feature tokenization transformer (SSFTT) [22], and local semantic feature aggregation-based transformer (LSFAT) [23] are compared. These models utilize the self-attention mechanism, which establishes relationships between features on a global scale.

The SSTN, SSFTT, and LSFAT models represent advanced architectures for feature extraction and classification, each combining convolutional and attention-based mechanisms. SSTN begins with 2-D convolutions (Conv2d) for feature extraction, enhanced by LeakyReLU and BatchNorm2d, and incorporates specialized 3-D convolutional layers (SPC32) and spatial attention modules (SARes) with Softmax-based attention weighting for feature recalibration. SSFTT integrates 3-D and 2-D convolutions to capture spatial–temporal features, followed by BatchNorm, ReLU, and Dropout for regularization, and employs a Transformer module with Residual layers, Attention mechanisms, and MLP-Blocks to model feature dependencies. LSFAT combines 3-D convolutions with transformer-based modules, starting with a 3-D convolutional layer and projecting features into higher dimensional spaces via linear transformations (patch-to-embedding). Its core consists of multiscale transformer blocks with residual connections, self-attention, and MLP layers, using GELU activation and spatial reduction via strided convolutions. All models utilize Dropout for regularization, with LSFAT uniquely employing parallel linear layers for multiscale classification. Together, these architectures demonstrate the integration of convolutional operations, attention mechanisms, and transformer-based modules for robust feature learning and classification.

*d) Graph neural network:* SOTA methods based on GNNs: CEGCN [36] and hypergraph convolutional network with multiple hyperedges fusion (HGCN-MHF) [38] are compared. CEGCN combines CNNs for pixel-level spectral–spatial feature extraction and GCNs for superpixel-level contextual modeling, using a graph encoder–decoder to bridge Euclidean and non-Euclidean data structures, and learns adaptive edge weights. HGCN-MHF combines a CNN branch for spatial–spectral feature extraction and an HGCN branch with multiple hyperedges fusion for superpixel-level feature representation.

All models used PCA [44] for data preprocessing, reducing the dimensionality to 30 dimensions. Except for the SVM, all models adopted a patch size of $15 \times 15$. Specifically, for the Houston 2018 dataset, PCA was reduced to 25 dimensions, and the patch size selected was $9 \times 9$. The training epochs were set to 100. The equipment is equipped with a 12th Gen Intel Core i9-12900KF CPU, with a benchmark speed of 3.19 GHz, comprising 16 cores and 24 logical processors; it is also equipped with an NVIDIA GeForce RTX 3090 Ti graphics card, with a dedicated GPU memory of 24 GB; the memory capacity is 64 GB. All experiments were conducted on the software platforms PyTorch and PyTorch-Geometric.

### C. Preliminary Experiments on Unsupervised Learning

*1) Phased Training or End-to-End Training:* SGMAE is based on phased training. Phased training has many advantages. It allows feature extractors to learn more discriminative and invariant feature representations through SLL, contrastive learning, or pretraining on large-scale datasets, enhancing generalization ability while alleviating gradient problems in DNNs and improving robustness, [49]. In addition, the phased training decouples feature extraction from classifier training, reduces complexity, and improves training stability and efficiency, especially for small-scale datasets. The pretrained feature extractor can be reused for multiple downstream tasks, saving resources, [50]. However, phased training also has disadvantages such as insufficient task adaptability, limited joint optimization capabilities, dependence on pretraining task design, and high computational and time costs. For example, a pretrained feature extractor may not be able to fully adapt to the needs of a specific task and cannot respond to the optimization needs of the classifier, resulting in suboptimal solutions, [51]. At the same time, the design of pretraining tasks is crucial to the effectiveness of phased training, which increases the complexity of model development, [52]. On the contrary, these advantages are inherent in end-to-end networks. Therefore, it will be proved many times in this chapter that the performance of the proposed SGMAE is superior to that of a simple end-to-end network.

*2) Visualization on Unsupervised Learning:* SSL consists of an unsupervised feature extractor and a downstream supervised classifier. To explain the advantages of SGMAE in the unsupervised feature extraction stage, the extracted HSI features are visualized. The visualization experiment compared various unsupervised feature extraction methods and performed t-SNE [53] visualization on the features extracted without labels. The methods selected include $K$-Means [54], RCN [55], HSIMAE [56], and our proposed SGMAE. Fig. 4 shows the visualization obtained by extracting features from 300 samples excluding the background from the Indian Pines, Pavia University, and Salinas datasets, which were then reduced to 2-D using t-SNE and projected onto a Cartesian coordinate system. Fig. 5 illustrates the visualization of the entire data samples from these three datasets after feature extraction using multiple methods, which were then reduced to 3-D using t-SNE to simulate RGB images. The SGMAE displays the features extracted from each layer of the MGCNE, showing the output features from the first layer to the fourth layer, and the concatenation of these four layers' features. The visualization image demonstrates that the proposed SGMAE possesses strong multiscale feature extraction capabilities and can achieve good clustering under label-free conditions, showing potential for enhancing downstream classification tasks. The features presented can be directly used for classification.

*3) Pretraining of SGMAE:* The unsupervised training of SGMAE is a process independent of the classifier training. According to the loss function designed, as shown in (12), the basic problem we train is a binary classification problem, that is, predicting whether an edge exists. The AUC value and ROC curve can well reflect the classification performance of the model, so we set model validation at each epoch and calculate the AUC value and ROC curve of the current epoch, as shown in Fig. 6. It is learned that the model has good convergence ability on the HSI dataset with limited training epochs, and can be effectively applied to downstream multiclassification tasks.

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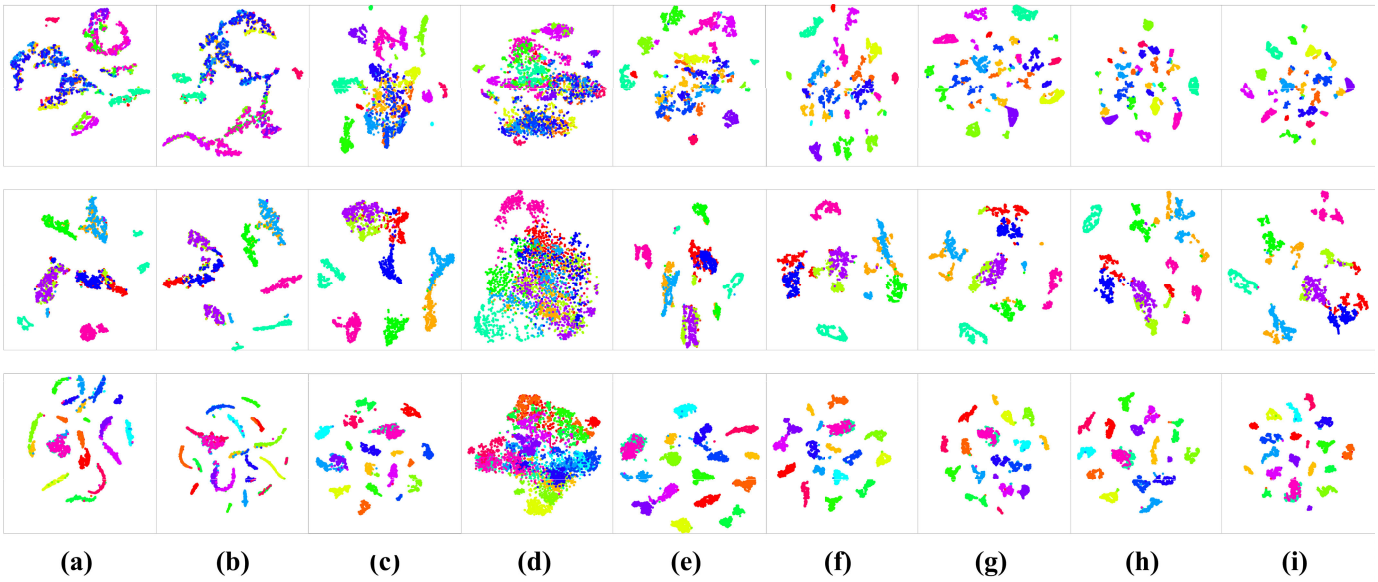



Fig. 4. Visualization results of the data after feature extraction using various self-supervised methods, reduced to a 2-D Cartesian coordinate system using t-SNE. From (top to bottom), each row corresponds to the three datasets: Indian Pines, Pavia University, and Salinas. (a) Original data. (b) $K$-means. (c) RCN. (d) HSIMAE. (e) SGMAE (the first GCN layer). (f) SGMAE (the second GCN layer). (g) SGMAE (the third GCN layer). (h) SGMAE (the fourth GCN layer). (i) SGMAE (concatenation of four GCN layers).

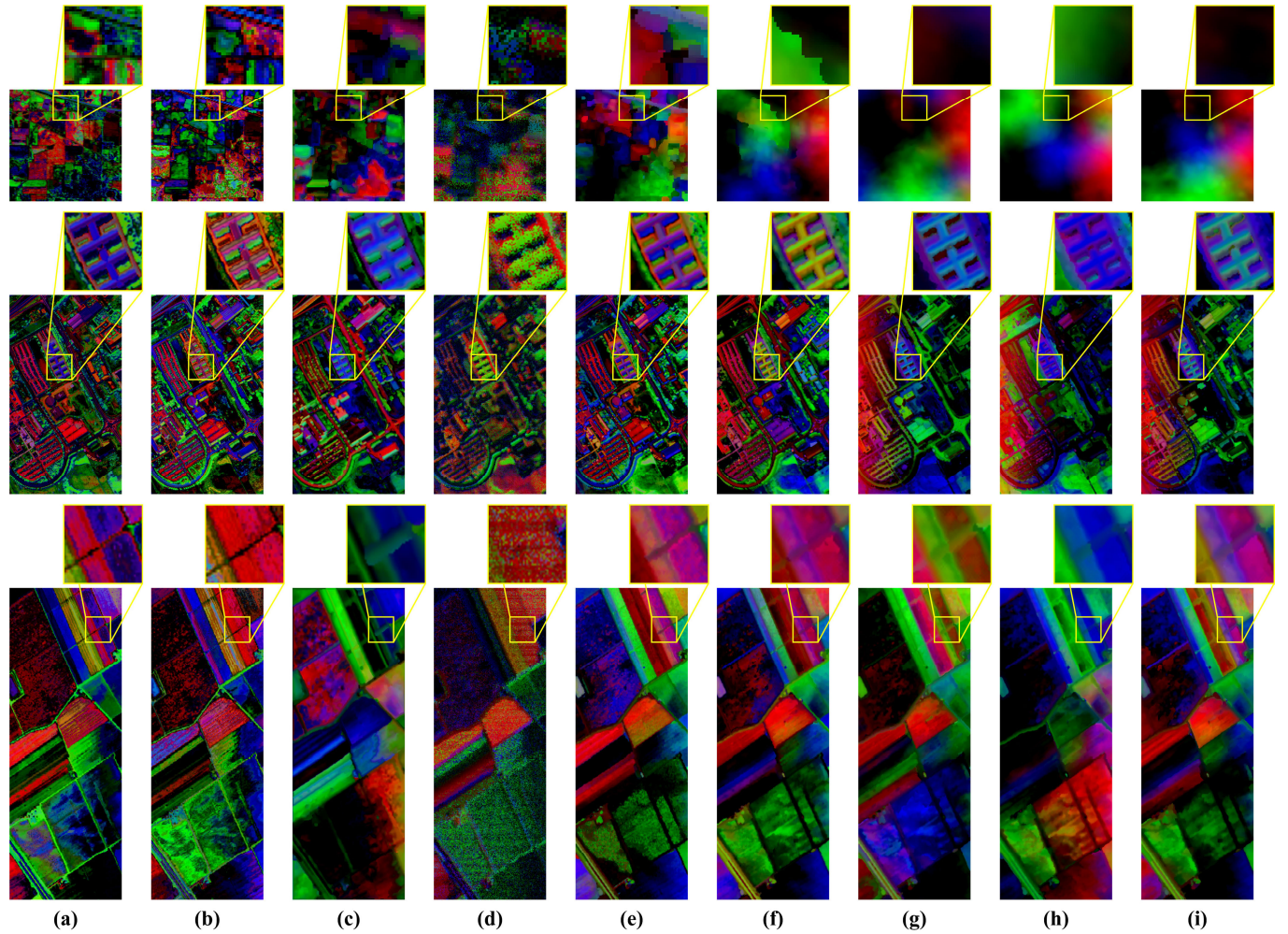



Fig. 5. Data after feature extraction using various self-supervised methods is reduced to three dimensions using t-SNE and simulated as RGB image feature maps. From (top to bottom), each row corresponds to the three datasets: Indian Pines, Pavia University, and Salinas. (a) Original data. (b) $K$-means. (c) RCN. (d) HSIMAE. (e) SGMAE (the first GCN layer). (f) SGMAE (the second GCN layer). (g) SGMAE (the third GCN layer). (h) SGMAE (the fourth GCN layer). (i) SGMAE (concatenation of four GCN layers).

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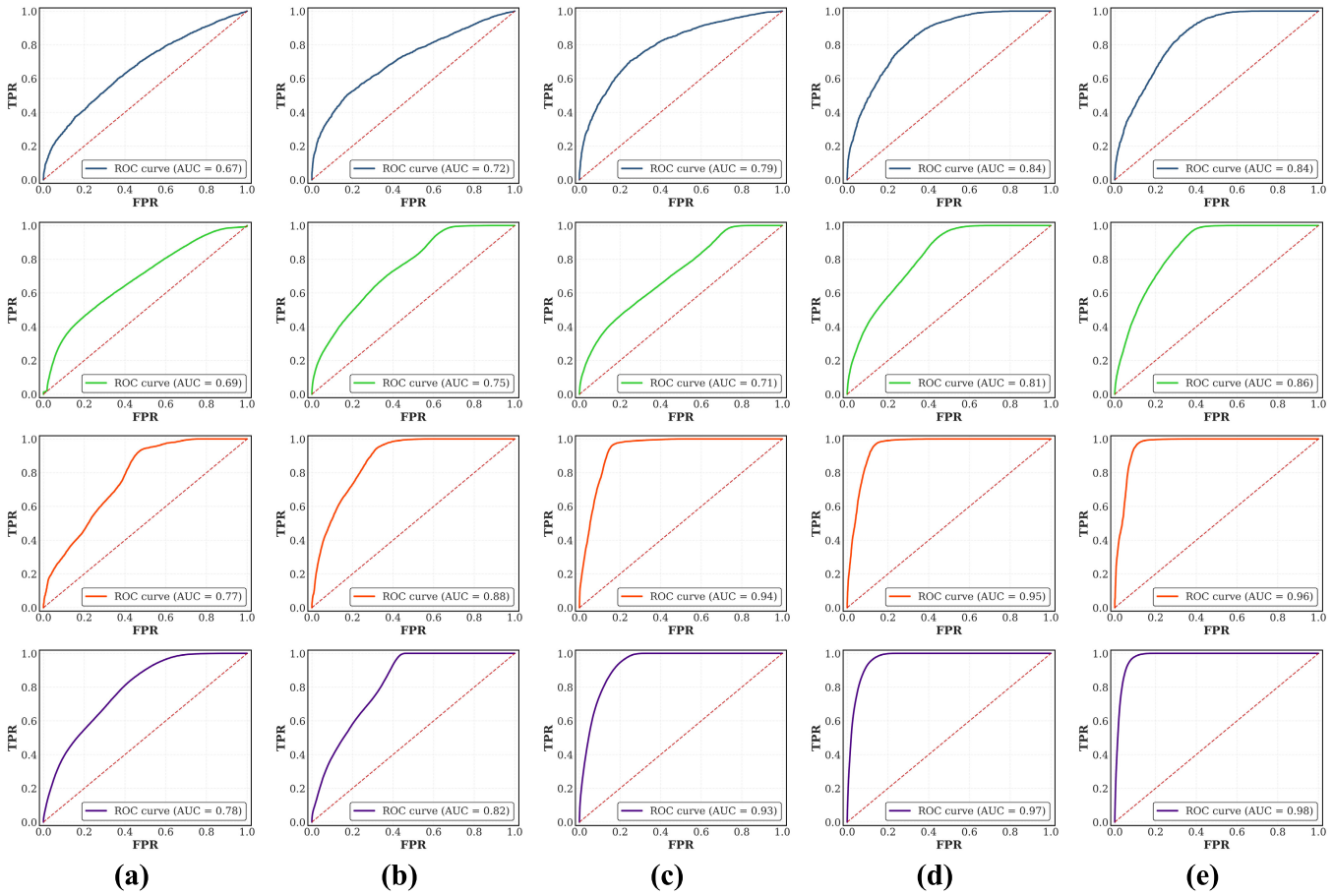



Fig. 6. ROC curve and AUC value for each epoch of evaluation during the unsupervised training of SGMAE. From (top to bottom), each row corresponds to the four datasets: Indian Pines, Pavia University, Salinas, and Houston 2018. (a) 2 epochs. (b) 4 epochs. (c) 6 epochs. (d) 8 epochs. (e) 10 epochs.

### D. Parameter Analysis

*1) Unified Setting:* Through preliminary experiments, the key parameters for further research on the proposed SGMAE and their appropriate default values were identified. The main hyperparameters include the number of training epochs, masking ratio, number of encoder layers, and decoder strategy, with default values set as follows: 10 epochs for training, a masking ratio of 0.6, three encoder layers, and the proposed CCD. All experimental data were averaged over ten repetitions to eliminate randomness. When studying a specific hyperparameter, it was treated as a variable while the other hyperparameters were set to their default values. All experiments' supervised learning training epochs are uniformly set to 100. The patch size for the Indian Pines, Pavia University, and Salinas datasets was set to 15, with PCA dimensionality reduction set to 30. For the Houston 2018 dataset, the patch size was set to 9, and the PCA dimensionality reduction was set to 25, with an adjacency strategy of four neighborhoods for edge construction.

*2) Training Epoch:* To save time for subsequent experiments, an appropriate number of training epochs that brings the model performance close to saturation needs to be found [see Fig. 7]. We conducted experiments on four datasets: Indian Pines, Pavia University, Salinas, and Houston 2018, setting a series of training epochs $\{1, 2, 3, 4, \ldots, 18, 19, 20\}$ and recording the OA of the model on the test set. It was found that the model's performance can improve with a smaller number of training epochs, and there is a probability of surpassing the average level after 10 epochs.

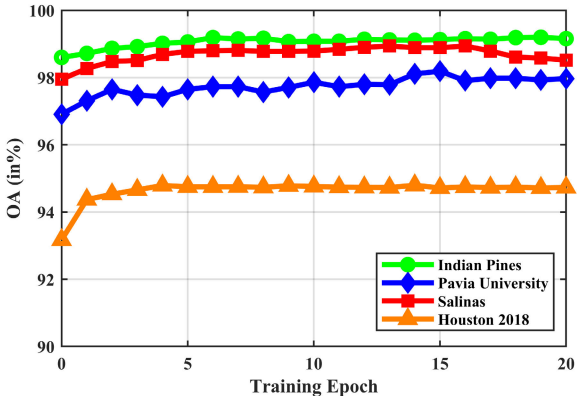



Fig. 7. Training Epoch. Set different maximum training epochs for the self-supervised SGMAE feature extraction model to evaluate the OA metrics of downstream classification tasks during from-scratch pretraining.

*3) Mask Ratio:* Fig. 8 illustrates the model's performance under different mask ratios across four datasets. On the Indian Pines and Houston 2018 datasets, the model demonstrates higher stability compared to Pavia University and Salinas, with small standard deviations in the results from ten repeated experiments. It is observed that the Indian Pines dataset is less affected by this parameter setting. The Pavia University

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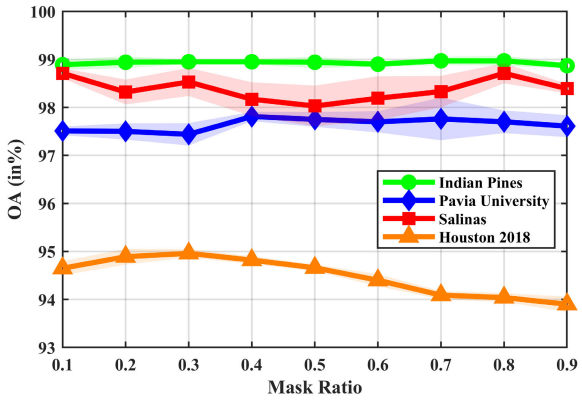



Fig. 8. Mask Ratios. Sensitivity analysis under different mask ratios.

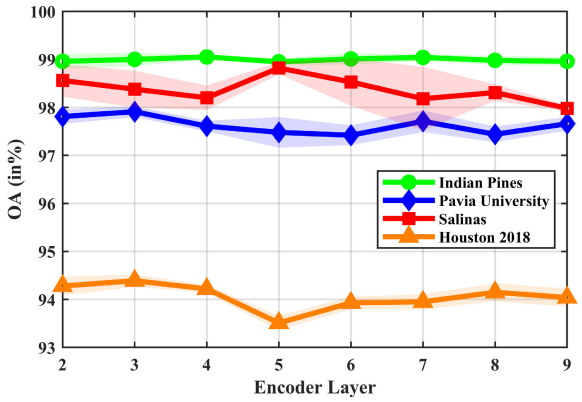



Fig. 9. Encoder layers. Sensitivity analysis under different encoder layers.

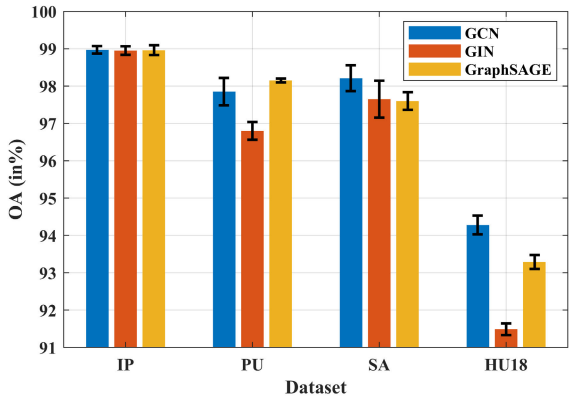



Fig. 10. Encoder strategy. Ablation analysis with different encoder strategies.

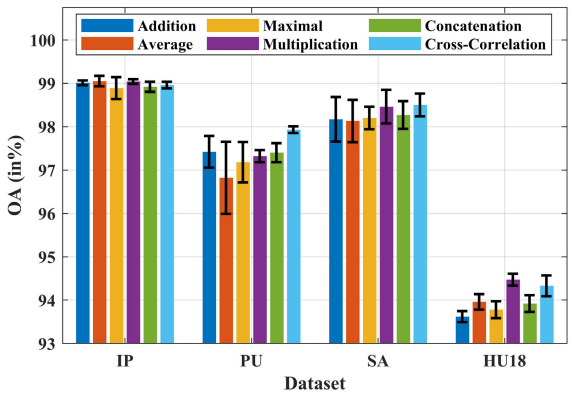



Fig. 11. Decoder strategies. Ablation analysis with different decoder strategies.

dataset performs better when the parameter setting is greater than 0.4. The Salinas dataset shows two peaks at parameter settings of 0.3 and 0.8, outperforming other settings. The Houston 2018 dataset exhibits optimal model performance at a parameter setting of 0.3, with performance decreasing as the setting deviates from 0.3. This indicates that the optimal mask ratio selection strategy is closely related to the dataset size and other characteristics.

*4) Encoder Layer:* Fig. 9 illustrates the model's performance under different Encoder Layer settings across four datasets, i.e., varying the number of GCN encoder layers. In addition, the computational cost of the decoder increases with the number of encoder layers. The Indian Pines, Pavia University, and Houston 2018 datasets exhibit higher model stability compared to the Salinas dataset, with smaller standard deviations in the repeated experimental results. The experimental results for the Indian Pines dataset are less affected by different parameter settings. The results for the Pavia University dataset do not show a clear pattern with different parameter settings, except for a peak performance when the parameter is set to 3. The Salinas dataset shows a maximum peak when the parameter is set to 5. The Houston 2018 dataset demonstrates a more obvious trend, with a significant dip when the parameter is set to 5, and maximum values when the parameter is set to 3 and 8, which are farther from 5.

## E. Ablation Analysis

*1) Encoder Strategy:* The encoder utilizes multilayer node neighborhood information interaction to achieve multiscale feature extraction. Three encoding strategies are compared: GCN [45], GIN [57], and GraphSAGE [58]. The results are shown in Fig. 10. It is found that GCN performs the best across all datasets, while GIN performs the worst on all datasets. GraphSAGE's performance is second only to GCN and slightly outperforms GCN on the Pavia University dataset.

GCN, GIN, and GraphSAGE are three GNN models with distinct features: GCN captures local structures via spectral graph theory and neighborhood aggregation, ideal for small-scale graphs; GIN enhances expressiveness through MLP and sum aggregation, excelling in graph classification; GraphSAGE samples and aggregates neighbor information for node embeddings, suited for large-scale graphs. In HSI classification, GCN outperforms due to its ability to effectively capture local spatial–spectral information, aligning with the stable graph structure of HSI data. GIN and GraphSAGE may overfocus on global structures or sampling, underutilizing local details. Thus, GCN balances local feature extraction and computational efficiency better, delivering superior performance.

*2) Decoder Strategy:* The decoder computes the source and destination node vectors of the masked edges to achieve their vector representation, thereby facilitating the fusion of multiscale feature information. Six decoding strategies are compared: element-wise addition, element-wise averaging, element-wise maximum, element-wise multiplication, concatenation along the spectral dimension, and the proposed cross correlation calculations. The results are shown in Fig. 11. The

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TABLE II

ENHANCEMENT FOR DOWNSTREAM CLASSIFIERS ABLATION ANALYSIS OF THE PROPOSED METHOD ON THE FOUR DATASETS

| Classifiers | MGCNE | CCD | Datasets | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Indian Pines | | | Pavia University | | | Salinas | | | Houston 2018 | | |
| | | | OA(%) | AA(%) | $\kappa\times100$ | OA(%) | AA(%) | $\kappa\times100$ | OA(%) | AA(%) | $\kappa\times100$ | OA(%) | AA(%) | $\kappa\times100$ |
| SVM | × | × | 79.87 | 72.77 | 76.91 | 85.18 | 80.45 | 80.03 | 86.19 | 90.45 | 84.58 | 78.26 | 62.81 | 71.12 |
| | ✓ | × | **96.23** | **88.67** | **95.68** | **91.08** | **83.52** | **87.74** | **92.88** | **94.76** | **92.05** | **92.38** | **89.25** | **90.83** |
| | ✓ | ✓ | **96.09** | **87.79** | **95.51** | **94.44** | **91.44** | **92.59** | **91.79** | **93.47** | **90.82** | **95.57** | **91.84** | **94.23** |
| CNN | × | × | 98.11 | 96.62 | 97.84 | 95.48 | 92.21 | 93.99 | 96.50 | 97.51 | 96.10 | 94.43 | 90.73 | 91.96 |
| | ✓ | × | **98.87** | **97.61** | **98.71** | **95.51** | 91.92 | **94.55** | 96.13 | **97.97** | 95.69 | **95.15** | **90.96** | **93.73** |
| | ✓ | ✓ | **99.01** | **98.52** | **98.87** | **97.13** | **94.06** | **96.19** | **96.63** | **97.68** | **96.25** | **95.73** | **92.87** | **94.44** |
| SSRN | × | × | 98.85 | 97.85 | 98.69 | 98.27 | 94.59 | 97.66 | 98.48 | 97.96 | 98.31 | 95.49 | 92.68 | 94.43 |
| | ✓ | × | **99.14** | **99.00** | **99.02** | 97.09 | 93.41 | 95.82 | 97.68 | 97.61 | 97.42 | 95.70 | 92.37 | 94.45 |
| | ✓ | ✓ | **98.89** | **98.05** | **98.73** | **98.71** | **94.87** | **97.69** | **98.65** | **97.98** | **98.42** | **96.62** | **93.00** | **95.56** |
| A2S2K | × | × | 99.00 | 97.85 | 98.86 | 98.20 | 96.05 | 97.61 | 97.74 | 97.49 | 97.48 | 95.49 | 92.61 | 94.43 |
| | ✓ | × | **99.06** | **98.28** | **98.93** | 97.89 | 95.30 | 96.88 | 96.62 | 97.13 | 96.24 | **96.55** | **93.28** | **95.48** |
| | ✓ | ✓ | **99.05** | **98.52** | **98.91** | **98.78** | **96.33** | **97.73** | **97.85** | **97.92** | **97.49** | **96.61** | **92.89** | **95.51** |
| SSTN | × | × | 98.84 | 97.12 | 98.67 | 96.86 | 93.83 | 95.83 | 97.34 | 98.32 | 97.04 | 91.06 | 84.49 | 88.34 |
| | ✓ | × | **98.93** | **98.46** | **98.77** | **97.02** | **94.16** | **96.05** | **97.46** | **98.62** | **97.18** | **94.88** | **88.59** | **93.34** |
| | ✓ | ✓ | **98.98** | **98.26** | **98.83** | **99.04** | **97.60** | **98.73** | **98.51** | **97.34** | **98.34** | **95.40** | **91.32** | **93.68** |
| SSFTT | × | × | 98.77 | 97.22 | 98.59 | 97.33 | 95.00 | 96.45 | 97.33 | 97.96 | 97.03 | 94.13 | 89.24 | 92.36 |
| | ✓ | × | **98.79** | **98.23** | **98.62** | 96.50 | 93.00 | 95.35 | 96.67 | **98.18** | 96.29 | **95.39** | **90.31** | **94.01** |
| | ✓ | ✓ | **98.88** | **97.72** | **98.72** | **98.43** | **96.88** | **97.92** | **97.57** | **98.64** | **97.29** | **95.67** | **91.05** | **94.38** |

TABLE III

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\kappa$, AND THE ACCURACIES OF EACH CLASSES FOR INDIAN PINES DATASET

| Class | Conventional | | Residual Network | | Transformer | | | Graph Neural Network | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | SVM | CNN | SSRN | A2S2K | SSTN | SSFTT | LSFAT | CEGCN | HGCN-MHF | SGMAE |
| 1 | 44.39±11.01 | 99.13±1.44 | 99.57±1.30 | 96.96±3.40 | 98.48±1.96 | 99.78±0.65 | 96.74±5.17 | 95.12±3.78 | 97.32±2.03 | **100.00±0.00** |
| 2 | 74.25±3.35 | 95.18±0.35 | 96.48±0.52 | 97.01±0.25 | 96.47±0.24 | 96.57±0.43 | 96.94±0.35 | 98.34±0.81 | **98.48±0.56** | 97.34±0.36 |
| 3 | 59.68±2.84 | 99.08±0.58 | **99.84±0.14** | 99.69±0.25 | 99.69±0.08 | 99.00±0.64 | 99.52±0.40 | 98.36±1.80 | 98.07±1.68 | 99.73±0.19 |
| 4 | 52.25±5.04 | 98.44±0.78 | 96.54±8.10 | 99.49±0.65 | 98.90±0.39 | **99.96±0.13** | 98.52±1.23 | 98.93±1.18 | 97.71±1.93 | 99.75±0.51 |
| 5 | 88.14±2.47 | 99.11±0.37 | **99.54±0.46** | 99.28±0.45 | 98.57±0.25 | 99.67±0.30 | 98.80±0.93 | 96.90±2.20 | 98.74±1.34 | 99.38±0.86 |
| 6 | 96.41±1.27 | 98.29±0.38 | 99.68±0.23 | 99.42±0.17 | 99.45±0.22 | **99.82±0.25** | 99.30±0.53 | 99.67±0.25 | 99.58±0.47 | 99.75±0.05 |
| 7 | 60.80±14.95 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | 99.20±1.60 | 96.40±3.77 | **100.00±0.00** |
| 8 | 96.21±1.87 | 99.98±0.06 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | 99.94±0.19 | **100.00±0.00** | 99.82±0.48 | **100.00±0.00** | **100.00±0.00** |
| 9 | 37.78±18.56 | 77.00±5.57 | 84.00±18.95 | 87.00±6.00 | 81.00±13.00 | 83.50±14.33 | 89.00±8.31 | 80.56±15.57 | 92.78±9.64 | **96.00±3.74** |
| 10 | 72.93±3.85 | 98.18±0.44 | 98.80±1.55 | **99.37±0.29** | 99.16±0.13 | 97.98±0.77 | 98.07±2.55 | 98.04±1.13 | 98.35±1.16 | 98.85±0.29 |
| 11 | 84.51±1.70 | 98.97±0.24 | 99.14±0.55 | 99.42±0.07 | 99.59±0.15 | 99.62±0.14 | 99.49±0.26 | 99.46±0.49 | 99.53±0.33 | **99.71±0.08** |
| 12 | 64.25±3.47 | 94.38±0.74 | 98.38±0.69 | 98.21±0.30 | 98.08±0.36 | 97.86±0.79 | 97.30±0.73 | 98.85±0.78 | **98.95±0.84** | 98.28±0.76 |
| 13 | 96.27±3.13 | 99.07±0.46 | 98.78±2.71 | 98.20±1.09 | 97.61±0.41 | **99.90±0.29** | 99.61±0.43 | 99.78±0.49 | 99.51±0.38 | 97.66±4.68 |
| 14 | 96.50±0.95 | 99.93±0.09 | **100.00±0.00** | 99.96±0.06 | 99.94±0.07 | 99.94±0.08 | 99.91±0.16 | 99.78±0.17 | 99.90±0.12 | 99.72±0.27 |
| 15 | 54.09±3.77 | 99.02±0.64 | **99.79±0.25** | 99.53±0.47 | 99.64±0.13 | 99.02±1.03 | 99.09±1.41 | 99.06±1.60 | 98.94±1.85 | 99.69±0.30 |
| 16 | 85.95±1.17 | 90.11±4.80 | 95.05±4.72 | 92.04±4.11 | 87.31±4.30 | 82.90±5.82 | 96.02±4.33 | **99.29±0.95** | 96.30±2.63 | 93.98±1.46 |
| OA (%) | 79.87±0.47 | 98.11±0.13 | 98.85±0.54 | 99.00±0.12 | 98.84±0.08 | 98.77±0.17 | 98.81±0.38 | 98.91±0.31 | 99.03±0.26 | **99.12±0.26** |
| AA (%) | 72.77±2.06 | 96.62±0.29 | 97.85±1.90 | 97.85±0.77 | 97.12±0.78 | 97.22±1.00 | 98.02±0.74 | 97.57±1.08 | 98.16±0.84 | **98.74±0.53** |
| $\kappa\times100$ | 76.91±0.53 | 97.84±0.15 | 98.69±0.62 | 98.86±0.14 | 98.67±0.09 | 98.59±0.20 | 98.64±0.43 | 98.76±0.35 | 98.89±0.29 | **99.00±0.30** |

TABLE IV

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\kappa$, AND THE ACCURACIES OF EACH CLASSES FOR PAVIA UNIVERSITY DATASET

| Class | Conventional | | Residual Network | | Transformer | | | Graph Neural Network | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | SVM | CNN | SSRN | A2S2K | SSTN | SSFTT | LSFAT | CEGCN | HGCN-MHF | SGMAE |
| 1 | 84.67±1.57 | 95.93±0.69 | 99.27±0.70 | 99.29±0.65 | 96.96±0.35 | 97.22±0.99 | 96.98±0.51 | 98.90±1.36 | **99.33±0.92** | 98.48±0.66 |
| 2 | 95.30±1.08 | 99.38±0.19 | 99.91±0.03 | 99.97±0.07 | 99.95±0.05 | 99.77±0.44 | 99.84±0.11 | 99.96±0.05 | **99.97±0.05** | 99.64±0.08 |
| 3 | 54.53±6.08 | 80.00±2.94 | 95.77±3.99 | 93.92±2.33 | 77.99±2.57 | 88.53±6.41 | 87.06±3.53 | 92.84±9.05 | 96.07±6.05 | **98.21±0.72** |
| 4 | 80.59±5.32 | 83.31±2.37 | 89.62±2.49 | 90.81±1.10 | 91.59±2.31 | 89.06±7.16 | 91.09±1.97 | 95.12±1.69 | 92.68±5.17 | **98.42±0.87** |
| 5 | 86.18±11.82 | 97.32±3.24 | 99.20±0.56 | 99.24±0.92 | 99.41±0.28 | 98.74±1.72 | 99.56±0.26 | 99.96±0.09 | 99.97±0.07 | **100.00±0.00** |
| 6 | 65.72±3.21 | 99.16±0.36 | 99.96±0.11 | 99.92±0.19 | 98.63±0.57 | 99.82±0.28 | 98.13±0.65 | 99.59±0.95 | **100.00±0.00** | 99.03±1.66 |
| 7 | 76.53±6.58 | 99.35±0.41 | 99.53±0.89 | 99.60±0.40 | 99.11±0.94 | **99.81±0.56** | 98.26±0.99 | 99.18±1.35 | 98.49±2.76 | 98.54±0.76 |
| 8 | 82.10±5.42 | 89.12±2.02 | 98.89±0.51 | 95.72±2.19 | 94.87±1.51 | 94.83±1.68 | 93.91±2.04 | **98.99±1.01** | 98.09±1.33 | 97.74±1.26 |
| 9 | 98.47±0.87 | 86.34±5.56 | 78.19±7.00 | 85.97±8.80 | 85.95±3.86 | 87.20±6.77 | 92.05±2.95 | 98.99±1.16 | 94.60±4.97 | **99.54±0.55** |
| OA (%) | 85.18±0.70 | 95.48±0.19 | 98.27±0.37 | 98.20±0.28 | 96.86±0.34 | 97.33±1.20 | 97.20±0.32 | 98.93±0.45 | 98.83±0.48 | **99.04±0.28** |
| AA (%) | 80.45±1.87 | 92.21±0.60 | 95.59±1.11 | 96.05±1.00 | 93.83±0.70 | 95.00±2.27 | 95.21±0.48 | 98.17±0.96 | 97.69±1.00 | **98.84±0.36** |
| $\kappa\times100$ | 80.03±0.98 | 93.99±0.26 | 97.71±0.49 | 97.61±0.37 | 95.83±0.46 | 96.45±1.60 | 96.28±0.43 | 98.58±0.60 | 98.45±0.63 | **98.73±0.37** |

results show that cross correlation performs the best across all four datasets, as it achieves multiscale information fusion by first cross-combining vectors of multiple scales and then integrating them, working in conjunction with the encoder.

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TABLE V

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\kappa$, AND THE ACCURACIES OF EACH CLASSES FOR SALINAS DATASET

| Class | Conventional | | Residual Network | | Transformer | | | Graph Neural Network | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | SVM | CNN | SSRN | A2S2K | SSTN | SSFTT | LSFAT | CEGCN | HGCN-MHF | SGMAE |
| 1 | 97.44±1.22 | 97.70±1.64 | **100.00±0.00** | **100.00±0.00** | 99.71±0.30 | 99.96±0.09 | 99.96±0.10 | 99.73±0.49 | 99.17±1.19 | **100.00±0.00** |
| 2 | 99.22±0.67 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | 99.99±0.02 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** |
| 3 | 88.17±10.94 | 99.99±0.02 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | 99.88±0.33 | 99.92±0.18 | **100.00±0.00** | 99.63±1.11 | **100.00±0.00** |
| 4 | 97.84±1.40 | 96.13±0.65 | 98.51±0.67 | 91.87±10.10 | 98.77±0.54 | 98.15±0.47 | 96.45±1.31 | **99.14±1.51** | 98.83±2.14 | 98.67±0.90 |
| 5 | 93.60±3.19 | 99.48±0.42 | 99.03±0.47 | 99.72±0.16 | 99.75±0.19 | 99.81±0.38 | 99.44±0.62 | 97.76±2.98 | 99.10±0.55 | **100.00±0.00** |
| 6 | 98.62±1.36 | 99.69±0.18 | 99.72±0.19 | 99.94±0.05 | 99.89±0.08 | 99.78±0.21 | 99.91±0.18 | 99.90±0.21 | 99.84±0.15 | **99.97±0.03** |
| 7 | 99.32±0.17 | 98.36±0.77 | 99.93±0.07 | 99.89±0.16 | 99.96±0.04 | 98.95±1.14 | 99.48±0.87 | 99.98±0.03 | **99.99±0.03** | 99.94±0.00 |
| 8 | 80.16±7.25 | 93.37±0.42 | **98.93±0.47** | 96.78±1.13 | 91.44±1.17 | 95.62±0.72 | 92.81±0.99 | 95.88±2.64 | 95.84±2.64 | 98.51±0.34 |
| 9 | 98.92±0.42 | 99.59±0.21 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** |
| 10 | 85.47±3.51 | 98.44±0.48 | **99.37±0.18** | 99.21±0.40 | 99.35±0.36 | 98.87±0.68 | 98.19±0.91 | 96.57±3.48 | 95.73±3.21 | 99.30±0.15 |
| 11 | 84.40±7.77 | 96.17±1.27 | 99.04±0.63 | 93.55±5.91 | 99.32±0.60 | **99.67±0.22** | 99.43±0.52 | 99.04±1.39 | 98.35±3.64 | 99.11±0.70 |
| 12 | 98.48±1.70 | 99.97±0.06 | 99.77±0.15 | 99.82±0.12 | 99.78±0.12 | 99.33±0.53 | 99.80±0.38 | **100.00±0.00** | **100.00±0.00** | 99.95±0.05 |
| 13 | 97.50±2.08 | 98.78±0.61 | 84.31±6.39 | 92.51±5.26 | 94.50±3.73 | 93.96±2.64 | 95.47±1.60 | 99.23±1.04 | 97.96±3.81 | **99.84±0.16** |
| 14 | 88.89±2.00 | 95.52±2.14 | 94.28±2.30 | 93.65±2.57 | 94.92±1.83 | 93.18±2.56 | 95.37±3.33 | 97.93±1.91 | 97.68±5.38 | **99.02±0.70** |
| 15 | 51.19±8.55 | 89.89±1.07 | 94.53±1.28 | 93.22±1.91 | 95.91±0.77 | 90.64±1.32 | 90.85±1.27 | 97.86±1.72 | **98.32±1.61** | 94.27±0.76 |
| 16 | 87.98±6.05 | 97.06±1.32 | **99.88±0.21** | 99.75±0.22 | 99.78±0.09 | 99.65±0.33 | 99.71±0.28 | 97.87±2.24 | 99.42±1.43 | 99.72±0.28 |
| OA (%) | 86.19±1.07 | 96.50±0.24 | 98.48±0.20 | 97.74±0.41 | 97.34±0.21 | 97.33±0.47 | 96.80±0.28 | 98.35±0.73 | 98.39±0.78 | **98.79±0.12** |
| AA (%) | 90.45±1.11 | 97.51±0.27 | 97.96±0.48 | 97.49±0.76 | 98.32±0.29 | 97.96±0.47 | 97.92±0.25 | 98.81±0.33 | 98.74±0.46 | **99.27±0.06** |
| $\kappa \times 100$ | 84.58±1.18 | 96.10±0.26 | 98.31±0.22 | 97.48±0.46 | 97.04±0.23 | 97.03±0.52 | 96.44±0.31 | 98.17±0.81 | 98.20±0.87 | **98.65±0.14** |

TABLE VI

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\kappa$, AND THE ACCURACIES OF EACH CLASSES FOR HOUSTON 2018 DATASET

| Class | Conventional | | Residual Network | | Transformer | | | Graph Neural Network | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | SVM | CNN | SSRN | A2S2K | SSTN | SSFTT | LSFAT | CEGCN | HGCN-MHF | SGMAE |
| 1 | **89.13±0.53** | 82.66±5.46 | 86.81±0.76 | 85.25±2.21 | 85.20±2.48 | 75.16±3.00 | 74.55±3.34 | 87.77±1.23 | 86.38±1.65 | 85.91±5.99 |
| 2 | 94.62±0.25 | 94.89±1.79 | 93.68±1.06 | 96.17±1.34 | 92.86±2.47 | 94.26±0.88 | **97.54±1.41** | 95.87±0.61 | 96.59±0.56 | 96.02±1.28 |
| 3 | **100.00±0.00** | 98.78±0.28 | 98.27±0.54 | 91.39±0.99 | 96.83±0.15 | 99.13±0.45 | 96.78±1.89 | 99.74±0.32 | 57.87±47.30 | 99.85±0.13 |
| 4 | 94.36±0.09 | 97.83±2.00 | 98.30±0.47 | 96.87±0.66 | 95.43±1.92 | 96.28±0.29 | 92.36±2.06 | 97.75±0.30 | 97.71±0.24 | **98.44±0.85** |
| 5 | 43.78±6.66 | 88.56±1.07 | 94.22±0.21 | 91.91±0.46 | 81.24±1.86 | 88.12±2.03 | 83.85±1.27 | 91.51±2.05 | 91.05±1.51 | **96.03±1.12** |
| 6 | 75.42±4.79 | 99.27±0.31 | 96.72±0.29 | **99.84±0.13** | 98.21±0.84 | 99.64±0.25 | 99.69±0.08 | 99.01±0.55 | 86.42±24.49 | 99.62±0.52 |
| 7 | 67.17±1.18 | 98.28±2.36 | 97.16±1.90 | **99.62±0.13** | 78.53±11.22 | 78.07±4.60 | 92.86±1.69 | 81.54±9.45 | 0.00±0.00 | 98.76±1.20 |
| 8 | 81.68±0.63 | 95.58±0.90 | 97.35±0.88 | 97.98±0.71 | 91.86±0.58 | 96.24±0.54 | 93.74±1.18 | 98.73±0.16 | **98.89±0.37** | 97.43±0.78 |
| 9 | 93.69±0.10 | 98.72±0.20 | 95.42±0.48 | 93.33±0.11 | 98.31±0.14 | 98.69±0.13 | 97.61±0.88 | 99.01±0.06 | **99.28±0.17** | 98.90±0.21 |
| 10 | 41.21±0.99 | 85.79±1.42 | **92.98±1.11** | 90.34±1.95 | 78.69±2.09 | 83.29±3.60 | 78.02±1.68 | 88.43±0.81 | 91.09±0.78 | 90.76±1.60 |
| 11 | 50.11±0.76 | 76.29±3.21 | 86.56±0.61 | 85.23±0.99 | 70.19±3.64 | 80.40±1.67 | 75.45±0.39 | 84.00±0.41 | **86.94±1.59** | 86.91±0.97 |
| 12 | 0.00±0.00 | 31.24±3.36 | 40.66±4.05 | 43.81±3.19 | 13.32±2.70 | 30.52±1.50 | 20.88±2.14 | 34.51±2.87 | 9.47±12.13 | **54.62±4.44** |
| 13 | 66.43±1.87 | 92.51±0.12 | 93.49±0.08 | 95.54±0.83 | 86.54±0.79 | 93.07±1.62 | 82.86±0.26 | 95.59±0.24 | **97.04±1.01** | 94.78±1.01 |
| 14 | 52.92±0.20 | 94.20±1.61 | 94.34±1.12 | 96.21±1.22 | 83.15±4.71 | 90.78±2.05 | 85.90±0.09 | **97.02±0.85** | 96.67±2.07 | 97.45±0.42 |
| 15 | 76.25±3.08 | 99.02±0.20 | **99.84±0.30** | 99.78±0.17 | 96.90±0.19 | 98.68±0.78 | 98.57±0.31 | 99.43±0.31 | 99.56±0.25 | 99.56±0.19 |
| 16 | 50.80±2.79 | 93.78±1.76 | 97.38±0.85 | 97.94±0.78 | 92.21±1.24 | 96.25±1.15 | 93.96±0.82 | 97.60±0.34 | **98.13±0.58** | 97.80±0.58 |
| 17 | 0.00±0.00 | 96.68±2.73 | 96.68±1.06 | **100.00±0.00** | 48.75±7.67 | 91.55±2.45 | 80.14±9.25 | 86.20±8.57 | 36.00±30.71 | 98.63±1.73 |
| 18 | 38.72±2.84 | 93.73±1.75 | 96.49±0.48 | 90.95±1.58 | 86.22±8.85 | 91.23±2.12 | 85.00±3.07 | **96.83±0.52** | 96.51±1.36 | 95.33±1.63 |
| 19 | 40.10±2.18 | 98.18±0.07 | 95.62±0.36 | 97.97±0.21 | 92.15±2.83 | 95.88±1.19 | 91.17±1.84 | **99.10±0.91** | 77.84±36.33 | 98.59±0.60 |
| 20 | 91.16±1.36 | 99.85±0.05 | 99.69±0.01 | **99.99±0.01** | 99.40±0.17 | 99.69±0.13 | 99.82±0.09 | 99.90±0.13 | 81.70±36.26 | 99.95±0.07 |
| OA (%) | 78.26±0.29 | 94.43±0.69 | 95.49±0.06 | 95.49±0.47 | 91.06±0.06 | 94.13±0.21 | 91.23±0.22 | 95.88±0.09 | 95.83±0.98 | **96.13±0.11** |
| AA (%) | 62.81±0.39 | 90.73±0.38 | 92.68±0.05 | 92.61±0.91 | 84.49±1.78 | 89.24±0.91 | 86.04±0.69 | 91.48±0.54 | 79.26±4.38 | **94.15±0.32** |
| $\kappa \times 100$ | 71.12±0.41 | 91.96±0.84 | 94.43±0.11 | 94.43±0.55 | 88.34±0.10 | 92.36±0.30 | 88.57±0.28 | 94.65±0.11 | 94.55±1.31 | **94.96±0.14** |

*3) Enhancement for Downstream Classifiers:* We analyze the impact of the proposed SGMAE feature extraction module, including the MGCNE and the CCD, on performance improvements when applied to different mainstream DL-supervised classification models. For each classifier, three scenarios are compared: the case without the SGMAE feature extraction module (×|×), the case with the SGMAE feature extraction module but without the CCD (✓|×), and the case with both the MGCNE and CCD of the SGMAE feature extraction module (✓|✓). For the case without the CCD, element-wise multiplication of the two vectors as a substitute is used for the necessary decoder. The ablation experiments are shown in Table II, with bold font in the table representing the performance improvement brought about by the proposed SGMAE.

The main reason for the performance enhancement of SVM is its lack of spatial information interaction, which is addressed by the SGMAE feature extraction module that eliminates noise while retaining the advantages of SVM in classification accuracy. This leads to significant enhancements in OA of 16.22%, 9.26%, 5.6%, and 17.31% across the four datasets. For DL models, the proposed SGMAE consistently provides stable enhancements. Notably, the Houston 2018 dataset, which has a significantly larger sample size, exposes various shortcomings of many traditional methods. However, classifiers (such as SSTN) using our feature extraction module achieve substantial performance enhancements. Therefore, the SGMAE feature extraction pretrained module can enhance the performance of SVM and most DL classifiers in downstream classification

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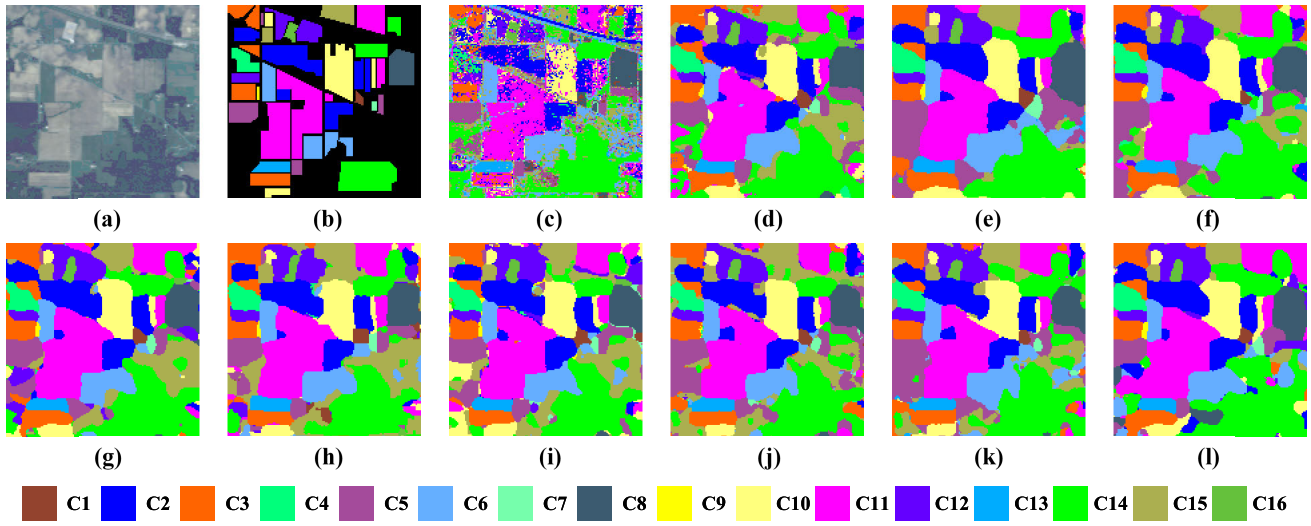



Fig. 12. Classification maps obtained by the compared methods on Indian Pines dataset. (a) False color map. (b) Ground truth. (c) SVM. (d) CNN. (e) SSRN. (f) A2S2K. (g) SSTN. (h) SSFTT. (i) LSFAT. (j) CEGCN. (k) HGCN-MHF. (l) SGMAE.

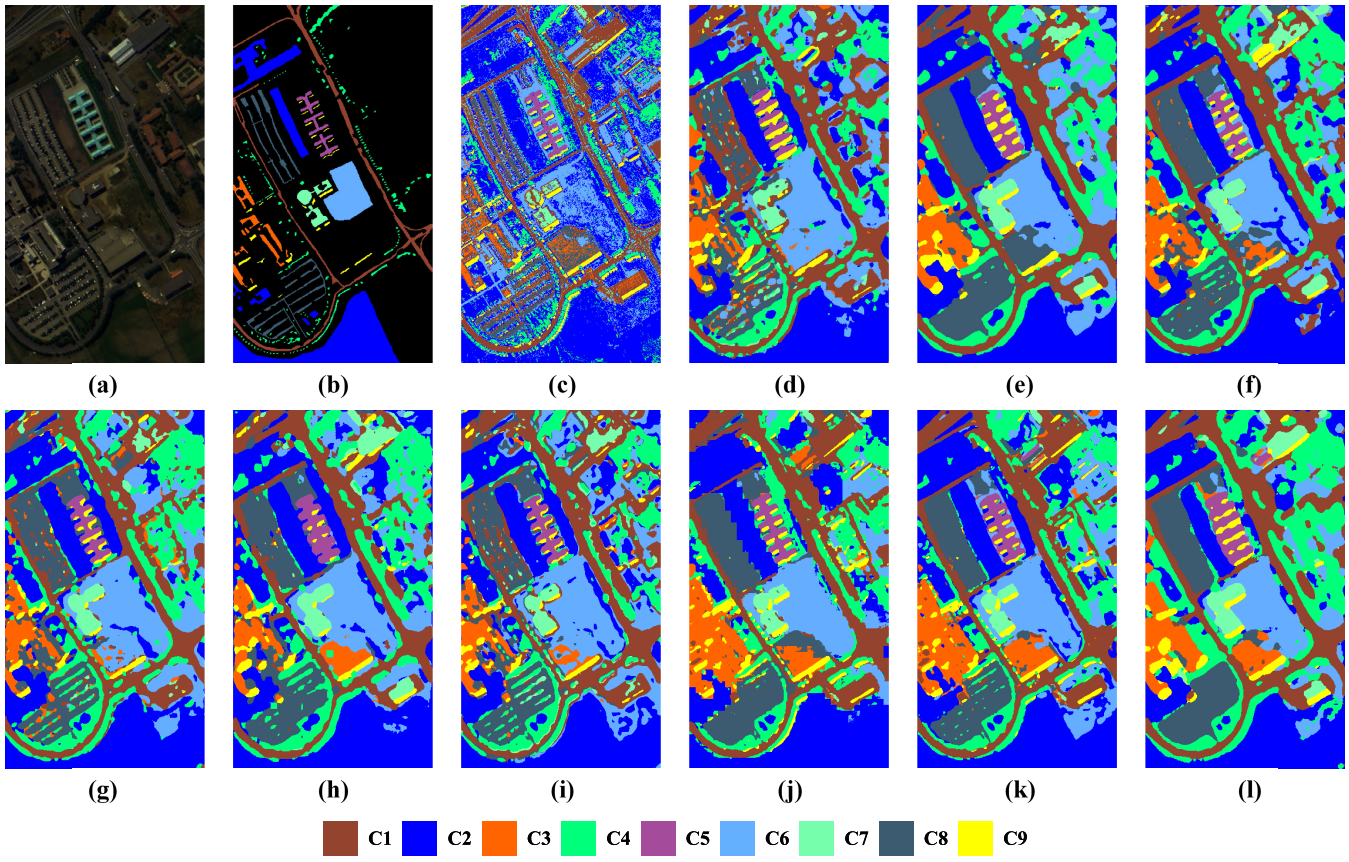



Fig. 13. Classification maps obtained by the compared methods on the Pavia University dataset. (a) False color map. (b) Ground truth. (c) SVM. (d) CNN. (e) SSRN. (f) A2S2K. (g) SSTN. (h) SSFTT. (i) LSFAT. (j) CEGCN. (k) HGCN-MHF. (l) SGMAE.

tasks. The enhancement effect of the CCD has also been validated across most datasets and methods.

### *F. Experimental Comparison With Competitive Approaches*

In this section, we conduct a quantitative analysis of the classification performance of the comparison methods through classification accuracy and visualization charts. The classification results of all methods on the Indian Pines, Pavia University, Salinas, and Houston 2018 datasets are presented in Tables III–VI and Figs. 12–15. To ensure fairness, the experiments were repeated ten times to obtain the OA, AA, $\kappa$ statistic, as well as the mean and standard deviation of the accuracy for each class.

On the four datasets–Indian Pines, Pavia University, Salinas, and Houston 2018–our model leads in all three metrics: OA, AA, and Kappa.

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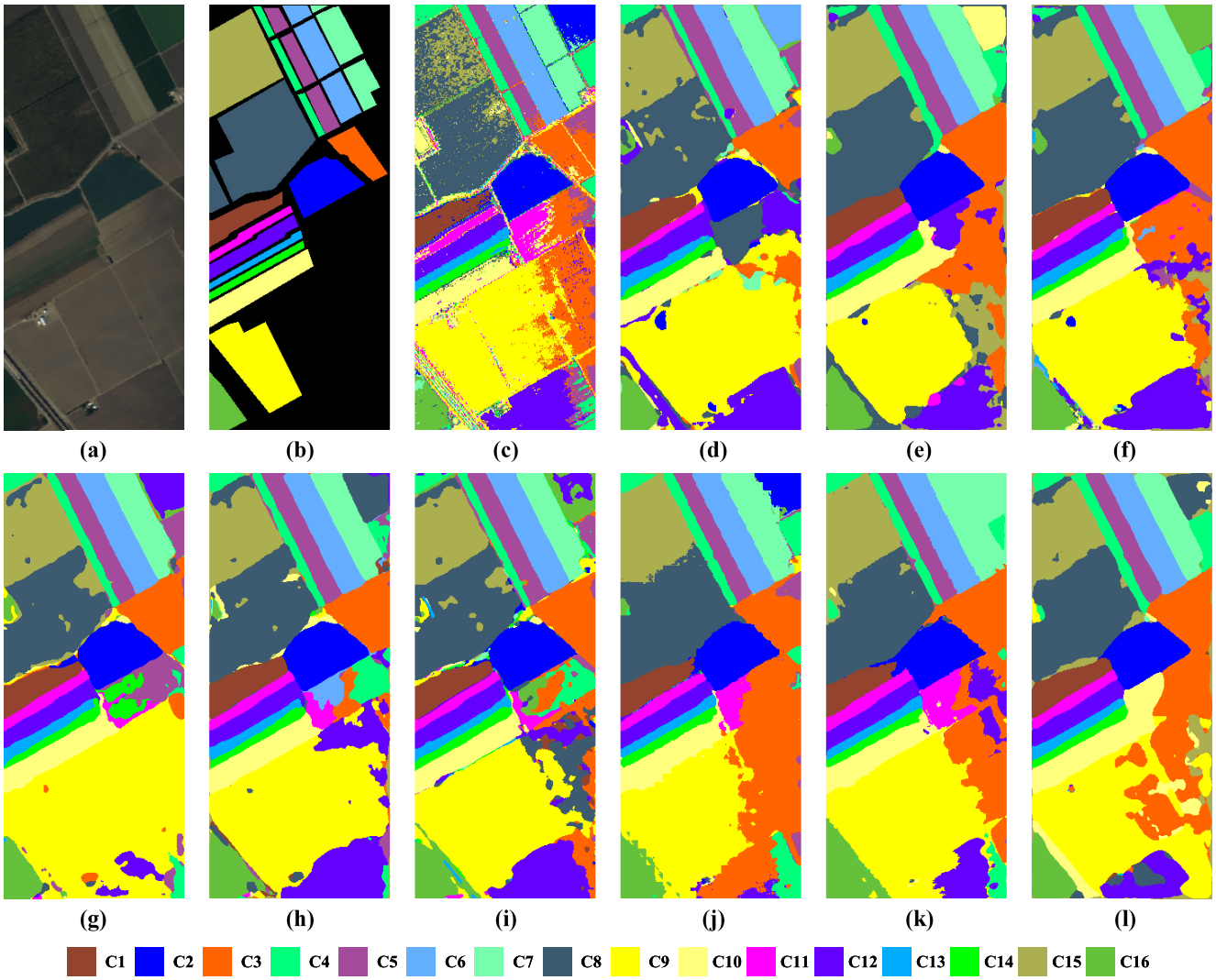



Fig. 14. Classification maps obtained by the compared methods on the Salinas dataset. (a) False color map. (b) Ground truth. (c) SVM. (d) CNN. (e) SSRN. (f) A2S2K. (g) SSTN. (h) SSFTT. (i) LSFAT. (j) CEGCN. (k) HGCN-MHF. (l) SGMAE.

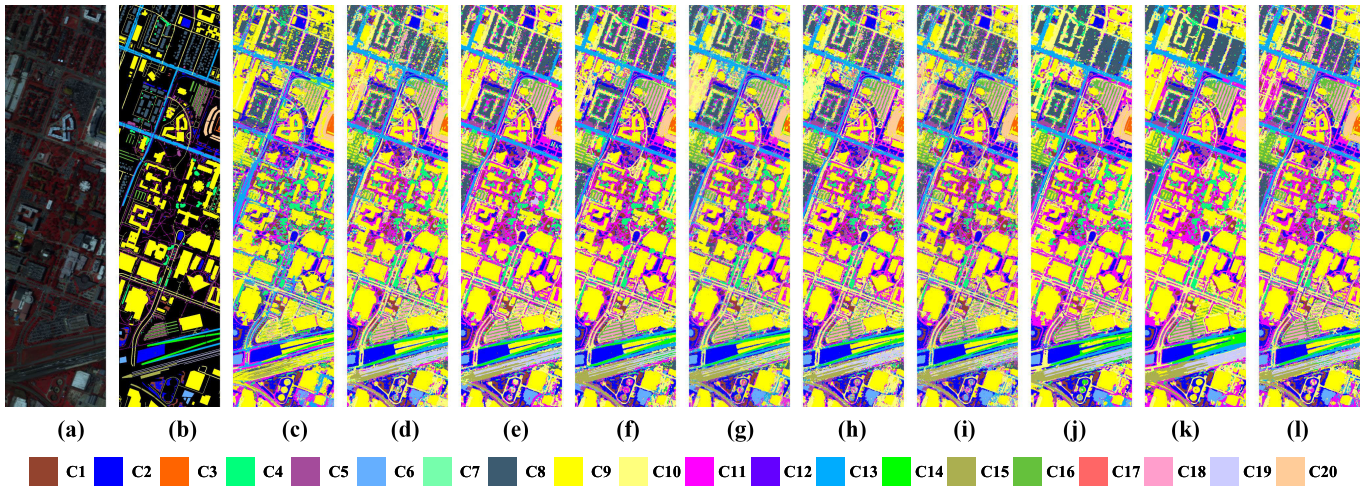



Fig. 15. Classification maps obtained by the compared methods on the Houston 2018 dataset. (a) False color map. (b) Ground truth. (c) SVM. (d) CNN. (e) SSRN. (f) A2S2K. (g) SSTN. (h) SSFTT. (i) LSFAT. (j) CEGCN. (k) HGCN-MHF. (l) SGMAE.

The SVM method, which does not use the patch approach, achieves lower accuracy with more noise but avoids edge blurring, making its results closer to the true state of false color images.

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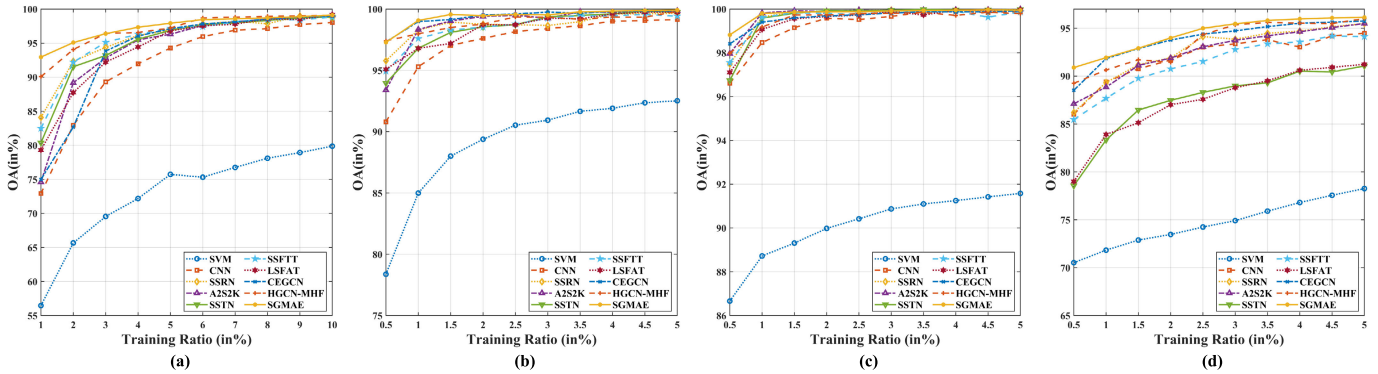



Fig. 16. Classification performance of the compared methods with different numbers of labeled data on four datasets. (a) Indian Pines. (b) Pavia University. (c) Salinas. (d) Houston 2018.

The CNN method, with its simple architecture, achieves reasonable accuracy at a low computational cost. Methods based on ResNet, with their deep layers, incur higher computational costs and exhibit the lowest stability, though they sometimes achieve very high accuracy, surpassing Transformer on the Pavia University, Salinas, and Houston 2018 datasets.

Transformer-based methods introduce optimized attention mechanisms, maintaining relatively good accuracy with low computational costs and high stability, making them perform well on small datasets.

GNN-based methods use superpixel segmentation to construct graphs, enhancing texture-level classification accuracy while maintaining high OA, though with higher time complexity compared to Transformer and conventional methods.

The proposed SGMAE constructs graphs at the pixel level, achieving even higher model accuracy. In addition, it employs a novel phased training approach, allowing the trained feature extractor to be combined with any independently trained classifier, flexibly and effectively further improving the accuracy of existing classifiers.

### G. Robustness Assessment

The classification performance of the proposed SGMAE and others was compared using various proportions of training samples randomly selected from four datasets. The training sample proportions for the Indian Pines dataset are $\{1.0\%, 2.0\%, \ldots, 10.0\%\}$, while those for the Pavia University, Salinas, and Houston 2018 datasets are $\{0.5\%, 1.0\%, \ldots, 5.0\%\}$. As shown in Fig. 16, the classification performance gradually improves with the increase in training sample proportions. Furthermore, the proposed SGMAE consistently outperforms other research methods in OA and demonstrates exceptional classification performance. The experimental results further indicate that this method demonstrates strong robustness, with stable classification outcomes.

## V. Conclusion and Future Work

In this work, an SGMAE model is proposed, which is a pretrained multiscale feature extractor for HSI classification. It achieves label-free feature extraction for the complete set of samples in HSI data and facilitates the extraction and fusion of multiscale spatial–spectral information. Specifically, an MGCNE is proposed to achieve efficient extraction of multiscale spatial–spectral features, enhancing the model's capability to handle complex scenarios. A CCD is introduced to effectively fuse and represent the extracted multiscale features, with the reconstructed edges being used to train the parameters of the autoencoder network under the contrastive learning loss function. The pretrained encoder possesses feature extraction capabilities, significantly improving model performance in downstream classification tasks and demonstrating strong generalization abilities.

In the future, we aim to assess the applicability of this feature extraction SGMAE in other tasks, such as hyperspectral change detection and anomaly detection, as well as its potential in signal fusion across various modalities due to its label-free nature.

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